Forecasting the Variance of the Closing Half Hour for 0DTE SPX Options

Bullet-point draft of the full paper

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Working draft, 15 October 2026 — not for citation

1 Introduction

- **Question.** At the forecast origin, 15:30 ET on trading day t, we forecast the realized variance of the S&P 500 over the closing half hour (15:30–16:00 ET), $\mathbb{E}_t[\mathrm{RV}_t]$, where $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\,\cdot \mid \mathcal{F}_t]$ and the information set $\mathcal{F}_t$ holds all data through 15:30 on day t.
  - $\mathrm{RV}_t = \sum_{k=1}^{N_{t,c}} r_{t,c,k}^2$: the sum of the squared one-minute log returns $r_{t,c,k}$ of E-mini futures over the closing interval $i = c$, with $N_{t,c} \le 30$ the number of one-minute returns (Andersen et al., 2003).
  - A zero-days-to-expiration (0DTE) SPX Weeklys (SPXW) at-the-money straddle position, approximately delta-neutral at entry, entered at 15:30 and held unhedged to expiration, settles at the 16:00 close: its payoff is set by the index move over exactly this half hour (Section 7.1).
  - Forecast and position:
$$\hat{\sigma}_t^2 = \hat{y}_t^2 \, s_{t,c}^2, \qquad q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2).$$
  - $y_t = \sqrt{\mathrm{RV}_t / s_{t,c}^2}$: periodicity-adjusted realized volatility. $s_{t,c}^2$: the intraday periodicity factor, the mean of RV in the closing interval over the previous 20 trading days (Andersen and Bollerslev, 1997). IV_t^2: the option-implied variance of the closing half hour, from the 15:30 at-the-money SPXW implied volatilities.
  - $q_t = +1$ buys the straddle, $q_t = -1$ sells it ($q_t = -1$ on ties); the strategy's daily return is $q_t R_t$, with R_t the straddle's simple return from 15:30 to settlement.
  - Sample: $T_{\mathrm{oos}} = 866$ out-of-sample trading days with an SPXW expiration, 3 January 2020 to 30 April 2024, early closes excluded.
- **Why it matters.**
  - Implied variance exceeds the forecast, $\hat{\sigma}_t^2 < \mathrm{IV}_t^2$, on 72–83% of trading days, depending on the model, in line with a positive variance risk premium; a forecast adds value mainly by identifying the days on which to buy.
  - 0DTE options account for a large share of S&P 500 index option volume (Bandi et al., 2023; Beckmeyer et al., 2023).
  - The payoff depends on a single, pre-specified window, so the forecast target and the holding period coincide.
- **Two specification choices, one forecast function:**
$$\hat{y}_t = f_{m,\mathcal{X},\mathcal{I}}(\boldsymbol{x}_t), \qquad \boldsymbol{x}_t \in \mathbb{R}^p.$$

  - Predictor set $\mathcal{X} \in \{\text{HAR}, \text{Full}\}$: the HAR benchmark, $p = 22$ (HAR components and calendar dummies), vs the full predictor set, $p = 628$ (adding 47 exogenous series in 8 groups).
  - Estimation sample $\mathcal{I} \in \{\{c\}, \{1, \ldots, M\}\}$: the closing-interval model (the closing half hour only, one observation per trading day) vs the pooled intraday model (every 30-minute interval, $M = 48$).
  - Model class m: linear (OLS, ridge, lasso, elastic net), tree ensemble (random forest, LightGBM, XGBoost) or neural network (MinGRU), crossed with $(\mathcal{X}, \mathcal{I})$.
- **Evaluation loss** (Patton, 2011):

$$\overline{\text{QLIKE}} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} L_t, \qquad L_t = \text{QLIKE}(\text{RV}_t, \hat{\sigma}_t^2), \qquad \text{QLIKE}(\text{RV}, \hat{\sigma}^2) = \frac{\text{RV}}{\hat{\sigma}^2} - \log \frac{\text{RV}}{\hat{\sigma}^2} - 1,$$

  with $\mathcal{T}_{\text{oos}}$ the set of out-of-sample trading days.
  - Economic evaluation: the annualized Sharpe ratio SR of $q_t R_t$ at mid-quotes.
- **Contributions.**
  - *Question.* We forecast the variance of the closing half hour, the risk a 0DTE SPX straddle opened at 15:30 carries to settlement.
  - *Data.* One predictor vector $\boldsymbol{x}_t$ holds, for each of 47 exogenous series, HAR components of its value, of its missing-data indicator and, for zero-inflated series, of an activity indicator; $T_{\text{oos}} = 866$ out-of-sample trading days.
  - *Method.* A rolling-window out-of-sample exercise, re-estimated daily, in which every hyperparameter is selected on data before the forecast origin. We measure what re-estimation frequency, hyperparameter re-selection and search budgets add out of sample. Forecasts are retransformed as $\hat{y}_t^2 s_{t,c}^2$, with no smearing or recalibration. The model classes include a MinGRU recurrent network (Feng et al., 2024), regularized like the linear models.
  - *Findings.* The full predictor set lowers $\overline{\text{QLIKE}}$ relative to the HAR benchmark in 38 of 51 pairs, 3 of them significantly at the 5% level. The closing-interval model is more accurate than the pooled intraday model; for ridge on the HAR benchmark, $\overline{\text{QLIKE}}$ changes by -12.0% (DM $= -3.37$).
  - *Economic value.* 112 of the 112 forecasts earn a higher SR than the unconditional short-straddle benchmark ($q_t \equiv -1$), 20 of them significantly at the 5% level. Accuracy and economic value are only weakly related across forecasts.
- **Predictor set: full predictor set vs HAR benchmark** (each pair differs only in $\mathcal{X}$).
  - $\Delta\overline{\text{QLIKE}} = \overline{\text{QLIKE}}_{\text{Full}} - \overline{\text{QLIKE}}_{\text{HAR}} < 0$ in 38 of 51 pairs; median relative change -1.7%.
  - 3 of these reductions are significant at the 5% level (Diebold–Mariano statistic with the Harvey–Leybourne–Newbold correction); 0 of the 13 increases are.
  - $\Delta\text{SR} > 0$ in 50 of 51 pairs; median $+0.54$; 4 significant at the 5% level (HAC t-statistic on the daily return differential).
  - All 20 forecasts that significantly beat the short-straddle benchmark ($q_t \equiv -1$) use the full predictor set.
- **Estimation sample: closing-interval vs pooled intraday model.**
  - The closing-interval model has the lower $\overline{\text{QLIKE}}$ wherever both are estimated.
  - The difference is significant at the 5% level on the HAR benchmark in every model class;

for ridge, $\overline{\text{QLIKE}}$ changes by -12.0% (DM $= -3.37$).
  - The difference in SR is not statistically significant.
- **Statistical accuracy and economic value are weakly related.**
  - Spearman rank correlation of $\overline{\text{QLIKE}}$ and SR across the 112 forecasts: -0.20 (negative: a lower loss goes with a higher SR).
  - Of the 21 single specification changes, none moves SR significantly.
- **Roadmap:** data (Section 3), predictors (Section 4), methods (Section 5), statistical results, first for the predictor set and then for the estimation sample (Section 6), the economic value of the straddle strategy (Section 7), variable importance (Section 8) and the conclusion (Section 9).

2 Related work

- **Realized variance and the HAR model:** $\mathbb{E}_t[\text{RV}_{t+1}] = \beta_0 + \beta_d \,\text{RV}_t^{(d)} + \beta_w \,\text{RV}_t^{(w)} + \beta_m \,\text{RV}_t^{(m)}$, with $\text{RV}_t^{(d)}$, $\text{RV}_t^{(w)}$ and $\text{RV}_t^{(m)}$ the average daily realized variance over the last 1, 5 and 22 trading days. Our HAR components $\bar{y}_t^{(k)}$ take the same averages of y over k intraday intervals.
  Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), Andersen et al. (2006).
  - Panel HAR, estimated jointly across assets, the cross-asset counterpart of our pooled intraday model, which pools the intraday intervals of one index: Bollerslev et al. (2018).
  - Barndorff-Nielsen and Shephard (2002) (realized variance).
- **Intraday periodicity:** volatility follows a U-shaped intraday pattern; the standard adjustment divides by a periodicity factor, here $y_{t,i} = \sqrt{\text{RV}_{t,i}/s_{t,i}^2}$.
  Andersen and Bollerslev (1997) (intraday periodicity), Engle and Gallo (2006) (intraday measures as indicators of daily volatility), Hizmeri and Bevilacqua (2025) (morning volatility uncertainty).
  - Taylor and Xu (1997) (periodicity factor), Engle and Sokalska (2012) (intraday volatility).
- **0DTE options and the variance risk premium:** the variance risk premium, implied minus expected realized variance, $\mathbb{E}_t^{\mathbb{Q}}[\text{RV}_t] - \mathbb{E}_t^{\mathbb{P}}[\text{RV}_t]$, is positive on average.
  Hizmeri and Bevilacqua (2025); option pricing and implied volatility Black and Scholes (1973), Merton (1973).
  - 0DTE options: Bandi et al. (2023), Beckmeyer et al. (2023), Brogaard et al. (2023), Dim et al. (2024), Vilkov (2026), Almeida et al. (2025); the variance risk premium: Carr and Wu (2009), Bollerslev et al. (2009).
- **Machine learning for volatility and returns:** a nonlinear $f(\boldsymbol{x}_t)$ of many predictors.
  Gu et al. (2020) (returns), Christensen et al. (2023) (realized volatility); gradient-boosted trees Ke et al. (2017) (LightGBM), Chen and Guestrin (2016) (XGBoost); hyperparameter optimization Akiba et al. (2019) (Optuna).
  - Breiman (2001) (random forests), Friedman (2001) (gradient boosting); realized volatility: Bucci (2020), Rahimikia and Poon (2020), Zhang et al. (2024); MinGRU Feng et al. (2024); PufferLib Suarez (2024).

- **Forecast evaluation:** QLIKE is robust to noise in the volatility proxy: with a conditionally unbiased proxy it ranks forecasts as the true conditional variance would, and $\arg\min_{v>0} \mathbb{E}_t[\text{QLIKE}(\text{RV}_t, v)] = \mathbb{E}_t[\text{RV}_t]$.
  Patton (2011) (robust loss functions), Diebold and Mariano (1995) with the small-sample correction of Harvey et al. (1997), Hansen et al. (2011) (model confidence set), Ehm et al. (2016) (Murphy diagrams, elementary scores), Mincer and Zarnowitz (1969) (forecast-efficiency regressions).
  - Hansen (2005) (superior predictive ability), White (2000) (reality check), Romano and Wolf (2005) (StepM), Newey and West (1987) (HAC variance).
- **Shrinkage with many weak signals:**

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda(\alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2}\|\boldsymbol{\beta}\|_2^2)$$

  (ridge $\alpha = 0$, lasso $\alpha = 1$); multi-penalty ridge gives each block of predictors its own penalty, $\lambda \sum_j \omega_j \beta_j^2$ with the penalty weight ω_j constant within a block.
  Giannone et al. (2021) (the illusion of sparsity), Shen and Xiu (2025) (weak signals), Kozak et al. (2020) (shrinking the cross-section), Hoerl and Kennard (1970) (ridge), Tibshirani (1996) (lasso), Zou and Hastie (2005) (elastic net), van de Wiel et al. (2021) (multi-penalty ridge), van Wieringen (2023).
  - Zou (2006) (adaptive lasso, penalty weights), Friedman et al. (2010) (elastic-net parametrization).
- **Variable importance:** Shapley additive explanations (SHAP) decompose each forecast additively, $\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}$.
  Lundberg and Lee (2017) (SHAP), Lundberg et al. (2018) (TreeSHAP), Aas et al. (2021) (dependent features).
- **Retransformation** (one sentence): a smearing estimate would correct $\hat{y}_t^2 s_{t,c}^2$ for retransformation bias; we do not.
  Duan (1983), Manning and Mullahy (2001).

3 Data

- **Sources** (Table 1).
  - **E-mini S&P 500 futures (ES),** nearly 24 hours a day, 1998–2024, sampled on a 30-minute grid. Trading day $t = 1, \ldots, T$; intraday interval $i = 1, \ldots, M$, the half-hour intervals of the day; the closing interval $i = c$ is the closing half hour (15:30–16:00 ET).
  - $r_{t,i,k}$: the k-th one-minute log return in interval i of day t, $k = 1, \ldots, N_{t,i}$, with $N_{t,i} \leq 30$. Realized variance (Andersen et al., 2003):

$$\text{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2.$$

  - Other realized measures, same interval:

$$\sum_k r_{t,i,k}, \quad \sum_k |r_{t,i,k}|, \quad \sum_k r_{t,i,k}^3, \quad \sum_k r_{t,i,k}^4, \quad \text{RS}_{t,i}^{+} = \sum_k r_{t,i,k}^2 \, \mathbf{1}\{r_{t,i,k} > 0\},$$

$$\text{BV}_{t,i} = \sum_{k \geq 2} |r_{t,i,k}| \, |r_{t,i,k-1}|, \qquad \text{AC}_{t,i} = \sum_{k \geq 2} r_{t,i,k} \, r_{t,i,k-1},$$

plus trading volume and the observation count $N_{t,i}$. The third and fourth sums are the realized skewness and kurtosis of Amaya et al. (2015) without their normalizations (division by $\mathrm{RV}_{t,i}^{3/2}/\sqrt{N_{t,i}}$ and by $\mathrm{RV}_{t,i}^{2}/N_{t,i}$). $\mathrm{RS}_{t,i}^{+}$ is the positive realized semivariance (Barndorff-Nielsen et al., 2010); $\mathrm{BV}_{t,i}$ is the bipower variation of Barndorff-Nielsen and Shephard (2004) without its $\pi/2$ scaling; $\mathrm{AC}_{t,i}$ is the first-order realized autocovariance.

- **Cross-section of index constituents,** equal- and value-weighted (EW, VW), 1993–2024: six return moments, turnover, order-flow imbalance (signed turnover) and bid-ask spreads. Order-flow imbalance, for each weighting ($\mathrm{OFI}^{\mathrm{EW}}$, $\mathrm{OFI}^{\mathrm{VW}}$):

$$\mathrm{OFI}_{t,i} = \frac{\mathrm{Buy}_{t,i} - \mathrm{Sell}_{t,i}}{\mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i}} \in [-1, 1], \qquad \text{missing if } \mathrm{Buy}_{t,i} + \mathrm{Sell}_{t,i} = 0,$$

  with $\mathrm{Buy}_{t,i}$ and $\mathrm{Sell}_{t,i}$ the buy and sell turnover in interval i of day t.
- **StockTwits** attention and sentiment measures, from 2009 and 2010.
- **Cboe** volatility indices: VIX (from 2006), VIX3M (from 2009) and VVIX (from 2012), all to February 2024.
- Cboe option-volume series (the volatility-demand group), 2011 to August 2023.
- **FOMC** announcement times, 1993 to November 2023.
- **SPXW option quotes on expiration days** (S&P 500 Weeklys), every 30 minutes: bid, ask, midquote, underlying level, implied volatility and delta. 1,291 expirations, 2020–2025.
- *[name the vendor of each file].*

- **Intraday grid and cleaning.**
  - Grid: 30-minute intervals from the first observed realized variance. Interval i of day t is retained if and only if its realized variance is observed and it lies outside the weekend break:

$$(t,i) \in \mathcal{G} \iff \mathrm{RV}_{t,i} \text{ observed and } (t,i) \notin (\text{Fri } 20{:}00,\ \text{Sun } 18{:}30),$$

    with $\mathcal{G}$ the interval grid, the set of retained intervals.
  - Lags run backward through the retained intervals in calendar order, across days and overnight: $(t, i-j)$ denotes the j-th retained interval before (t,i), possibly on an earlier trading day.
  - Exchange holidays and early-close days remain in the futures sample; the option-trading sample excludes early-close days.
  - Exogenous series u: availability (missing-data) indicator, recorded before any filling,

$$a_{t,i} = \mathbf{1}\{u \text{ observed in interval } (t,i)\} \in \{0,1\}.$$

  - Then $u_{t,i}$ is forward-filled (last observation carried forward), without a limit on staleness. Every interval with $a_{t,i} = 0$ is later imputed with the full-sample median (Section 4, step 4).
  - Intraday periodicity factor (Andersen and Bollerslev, 1997). $\mathcal{D}_{t,i}$: the last 20 trading days $t' < t$ on which interval i is retained.

$$s_{t,i}^{2} = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i}, \qquad |\mathcal{D}_{t,i}| \geq 5.$$

  - $|\mathcal{D}_{t,i}| < 5$: $s_{t,i}^{2} = 1$, and the interval is excluded from evaluation.

– Target, the periodicity-adjusted realized volatility, and its winsorized version:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s^2_{t,i}}, \qquad \tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ Q_{0.01}(\mathcal{Y}_{t,i}),\ Q_{0.99}(\mathcal{Y}_{t,i})),$$

$$\mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, 240\},$$

with $Q_{0.01}(\cdot)$ and $Q_{0.99}(\cdot)$ the empirical 1st and 99th percentiles.

– Winsorization window: the 240 preceding intervals of all times of day, excluding the current interval.

- **Samples** (Table 2).
  – Pooled intraday sample: 300,317 half-hour intervals from June 1998, with a rolling estimation window of $W = 24{,}000$ intervals:

$$300{,}317 = 303{,}442 \text{ intervals on the grid} - 3{,}125 \text{ burn-in intervals.}$$

  – Closing-interval sample ($i = c$): 3,469 trading days, July 2010 to April 2024, with a rolling estimation window of $W = 2{,}000$ trading days.
  – 1,469 rolling-window forecasts from June 2018:

$$1{,}469 = 3{,}469 - 2{,}000.$$

  – Hyperparameter selection for the linear models on a pre-sample: closing-interval observations from 13 April 2004; validation samples from 6 January 2016 to 2 January 2020.
  – Out-of-sample period: 3 January 2020 to 30 April 2024.

$$T_{\text{oos}} = 866 \text{ trading days} = 872 \text{ SPXW expirations} - 6 \text{ early-close days.}$$

  – The 413 later expirations are held out.

4 Predictors

- **Eight groups of exogenous predictors, 47 series:** realized moments (7), liquidity (14), equal-weighted market (6), value-weighted market (6), sentiment (3), implied volatility (3), volatility demand (4), FOMC (4); listed in Appendix Table 30.

$$47 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 4.$$

  – FOMC:

$$\mathbf{1}\{\text{announcement in } (t,i)\}, \quad \mathbf{1}\{\text{announcement on day } t\}, \quad 1/(1+n^+_{t,i}), \quad 1/(1+n^-_{t,i}).$$

  – $n^{\pm}_{t,i}$: number of intervals to the next (+) and since the last (−) scheduled announcement.
- **Predictors built from each series.**
  – Trailing averages of the transformed series $\tilde{u}$ over horizons of k intervals, lagged one interval (HAR components; Corsi, 2009):

$$\bar{u}^{(k)}_{t,i} = \frac{1}{k}\sum_{j=1}^{k} \tilde{u}_{t,i-j}, \qquad k \in \mathcal{K},$$

$$\mathcal{K} = \begin{cases} \{1, 5, 25, 125, 625, 3125\} & \text{closing-interval model,} \\ \{2^0, 2^1, \ldots, 2^{11}\} \text{ (base 2, 12 horizons)} & \text{pooled intraday model.} \end{cases}$$

Table 1: Data sources.

Source	Series	First	Last	Obs.	Use
E-mini S&P 500 futures (ES), *[vendor]*	$RV_{t,i}$ and 9 other statistics of the one-minute log returns $r_{t,i,k}$ (moments, bipower variation, autocovariance, volume, return count)	1998-01-05	2024-04-30	303,444	Target, predictors
Index constituents, equal- and value-weighted, *[vendor]*	11 statistics per weighting: 6 realized moments, turnover (total, buyer- and seller-initiated), effective and quoted spreads	1993-01-04	2024-04-30	213,578	Predictors
StockTwits, *[vendor]*	Attention (message volume)	2009-03-23	2024-04-30	184,594	Predictors
	Sentiment score; sentiment-tagged messages	2010-06-01	2024-04-30	162,284	Predictors
Cboe, *[vendor]*	VIX	2006-05-22	2024-02-12	85,625	Predictors
	VIX3M	2009-08-12	2024-02-12	50,150	Predictors
	VVIX	2012-03-26	2024-02-12	38,508	Predictors
	Net volatility demand, 4 series: SPX or all index options; opening and closing, or opening trades only	2011-01-03	2023-08-31	54,062	Predictors
FOMC calendar, *[vendor]*	Scheduled announcement times, 14:30 ET	1993-02-03	2023-11-01	246	Predictors
SPXW options on expiration days, *[vendor]*	Bid, ask, midquote, underlying level, implied volatility, delta, time to expiration, early-close flag; 1,291 expirations	2020-01-03	2025-12-31	8,198,778	Straddle positions

Notes: Each series is recorded at the end of its 30-minute interval (ET) and merged on that interval-end time stamp; the SPXW quotes are time-stamped in UTC and loaded separately. First and Last: the first and last interval with an observation; Obs.: the number of such intervals before the intraday grid is formed (constituents: per weighting; FOMC: announcements; SPXW: quotes). The VIX indices, the volatility-demand series and the FOMC calendar end before the futures sample (2024-04-30); after their last observation the availability indicator $a_{t,i}$ is zero and the value is imputed (Sections 3 and 4). The data files also contain 6 other release indicators, an SPXW-implied variance and other VIX series, none of them used as a predictor. *[vendor]*: the vendor is not yet named in any data file or document (the option chain is attributed to OptionMetrics only in a code comment).

Table 2: Estimation and evaluation samples.

Sample	Start	End	Obs.	Window W
Intraday grid: 30-minute intervals with an observed $RV_{t,i}$, outside the weekend break	1998-01-05	2024-04-30	303,442	–
Panel A: Closing-interval models (trading days)				
Closing-interval observations, after the burn-in	2010-07-12	2024-04-30	3,469	2,000
Rolling-window forecasts	2018-06-25	2024-04-30	1,469	–
Hyperparameter-selection pre-sample (linear models)	2004-04-13	2020-01-02	3,947	2,000
Validation samples (4×250 days)	2016-01-06	2020-01-02	1,000	2,000
Panel B: Pooled intraday models (30-minute intervals)				
Estimation sample, after the 3,125-interval burn-in	1998-06-04	2024-04-30	300,317	24,000
Rolling-window forecasts	2001-07-13	2024-04-30	276,317	–
Tree hyperparameter pre-sample (before 2003)	1998-06-04	2002-12-31	36,285	24,000
Panel C: Out-of-sample period (trading days)				
All trading days (closing-interval forecasts)	2020-01-03	2024-04-30	1,087	–
Evaluation days (T_{oos}): SPXW expiration days (872) less early-close days (6)	2020-01-03	2024-04-30	866	–
Later SPXW expiration days, held out (413 full trading days)	2024-05-01	2025-12-31	419	–

Notes: Dates are interval-end dates (ET); Obs. and the rolling estimation window W are in the units of each panel. The burn-in, 3,125 intervals, is the longest HAR horizon and is applied to each series as built (the closing-interval series, the intraday grid and, for the linear pre-sample, the series of the two intervals of the last hour), so the closing-interval sample starts in 2010 and the pre-sample in 2004. The pooled linear models use $W = 96{,}000$ intervals (2,000 trading days); the tree pre-sample is re-estimated every 250 intervals. Each validation sample follows a 25-day gap and a 2,000-day estimation sample (Section 5.4). Full trading days: expiration days that are not early-close days.

  - k counts intervals of the full intraday grid, overnight included; the closing-interval model uses the rows $i = c$.
  - The same averages of the availability indicator, $\bar{a}_{t,i}^{(k)}$.
  - Averages of an activity indicator, $\bar{\chi}_{t,i}^{(k)}$, for 7 zero-inflated series.
  - Names in tables and figures: the bar and the interval index are dropped ($i = c$ in the closing-interval model), so $u_t^{(k)}$ is series u averaged over k intervals; e.g. $\mathrm{VIX}_t^{(25)}$ is the transformed VIX averaged over the 25 intervals ending at 15:30 on day t. Series symbols: VIX, VIX3M, VVIX; constituent aggregates carry EW or VW, as in $\mathrm{OFI}^{\mathrm{EW}}$, $\mathrm{RV}^{\mathrm{VW}}$, $\mathrm{Spread}^{\mathrm{VW}}$; FOMC. Indicator averages: $\bar{a}_t^{(k)}(u)$, $\bar{\chi}_t^{(k)}(u)$.

- **HAR benchmark:** the HAR components of the target (6 columns) and 16 calendar dummies $\boldsymbol{D}_{t,i}$ (day of the week, monthly option expiration, quarterly "quadruple witching" expiration, month end, quarter end and others).

$$\bar{y}_{t,i}^{(k)} = \frac{1}{k}\sum_{j=1}^{k} \tilde{y}_{t,i-j}, \qquad 16 = \underbrace{5}_{\text{day of week}} + \underbrace{4}_{\text{time of day}} + \underbrace{7}_{\text{expiration, month, quarter}}.$$

  - At the closing interval, four of the 16 dummies (hour, overnight, opening and closing interval) are constant, and the quadruple-witching closing-auction dummy equals the quadruple-witching dummy, so the column screen below drops these five at every estimation of the closing-interval model (Table 3).

- **Transformations, in order:**

  1. intraday periodicity adjustment, dividing by a factor estimated on the same intraday interval ($\mathcal{D}_{t,i}$ as for $s_{t,i}^2$) and bounded relative to its own trailing median:

$$\tilde{u}_{t,i} = \frac{u_{t,i}}{b_{t,i}^{(u)}}, \qquad b_{t,i}^{(u)} = \begin{cases} \mathrm{clip}\big(\mathrm{mean}_{t'\in\mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big), & u \geq 0, \\ \max\big\{\mathrm{clip}\big(\mathrm{sd}_{t'\in\mathcal{D}_{t,i}}\, u_{t',i},\ 0.5\,\rho_{t,i},\ 2\rho_{t,i}\big),\ 0.1\,\bar{\sigma}^{(u)}\big\}, & u \text{ signed}; \end{cases}$$

     - $\rho_{t,i}$: median of the 250 most recent unbounded factors of interval i (at least 20 required).
     - $\bar{\sigma}^{(u)}$: full-sample median of the standard deviation on the same intraday interval.
     - Factor of a nonnegative series zero or undefined: $b_{t,i}^{(u)} = 1$.
     - Not adjusted: VIX, VIX3M and VVIX, StockTwits sentiment, FOMC.
  2. a variance-stabilizing transformation by series type, $\tilde{u}_{t,i} \leftarrow g(\tilde{u}_{t,i})$, the first matching rule applying:

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, realized variance, turnover,} \\ & \text{bipower variation, effective spread,} \\ \mathrm{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \mathrm{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log\max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

     - Signed series with no type rule whose standard-deviation divisor sits at its floor on at least 50% of intervals (volatility demand) instead take an inverse hyperbolic sine

of the raw series:

$$\tilde{u}_{t,i} = \operatorname{asinh}(u_{t,i}/\kappa_{t,i}), \qquad \kappa_{t,i} = \max\{e_{t,i-1},\ 0.01\,\mathrm{med}_{j\geq 1}\, e_{t,i-j}\}.$$

  - $e_{t,i}$: exponentially weighted moving average of $|u|$ over intervals with $u \neq 0$, half-life 250 days.

3. winsorization at the rolling 1st and 99th percentiles, as for the target:

$$\tilde{u}_{t,i} \leftarrow \mathrm{clip}(\tilde{u}_{t,i},\ Q_{0.01}(\mathcal{U}_{t,i}),\ Q_{0.99}(\mathcal{U}_{t,i})), \qquad \mathcal{U}_{t,i} = \{\tilde{u}_{t,i-j} : j = 1, \ldots, 240\};$$

4. imputation of missing values by the full-sample median, kept alongside the availability indicator:

$$\tilde{u}_{t,i} \leftarrow \begin{cases} \tilde{u}_{t,i}, & a_{t,i} = 1, \\ \mathrm{med}\{\tilde{u}_{t',i'} : a_{t',i'} = 1\}, & a_{t,i} = 0; \end{cases}$$

5. activity indicators, for the 7 series whose full-sample mode accounts for at least 20% of intervals:

$$\chi_{t,i} = \mathbf{1}\{u_{t,i} \neq \mathrm{mode}(u)\};$$

6. trailing averages $\bar{u}^{(k)}_{t,i}$, $\bar{a}^{(k)}_{t,i}$, $\bar{\chi}^{(k)}_{t,i}$, $\bar{y}^{(k)}_{t,i}$;
7. robust standardization on the rolling estimation window (median and interquartile range, with floors), for the averages of series values only; written for the closing-interval model, whose rows t' are trading days ($i = c$ suppressed), with $\mathcal{W}_t = \{t - W, \ldots, t - 1\}$:

$$z^{(k)}_t = \frac{\bar{u}^{(k)}_t - \mathrm{med}_{t'\in\mathcal{W}_t}\, \bar{u}^{(k)}_{t'}}{\max\{\mathrm{IQR}^{(k)}_t,\ 0.01\max_{t'\leq t}\mathrm{IQR}^{(k)}_{t'},\ \mathrm{IQR}^{*(k)}_t\}}, \qquad \mathrm{IQR}^{(k)}_t = \mathrm{IQR}\{\bar{u}^{(k)}_{t'} : t' \in \mathcal{W}_t\}.$$

  - $\mathrm{IQR}^{*(k)}_t$: IQR of $\bar{u}^{(k)}_{t'}$, $t' < t$, over rows whose k-interval averaging window is fully observed.
  - Column whose mode accounts for at least 20% of rows: $\mathrm{IQR}^{*(k)}_t$ also excludes rows at the mode.
  - $\mathrm{IQR}^{*(k)}_t = 1$ until 8 such rows exist.
  - $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$: not standardized. HAR components, calendar dummies: $\mathrm{IQR}^{*(k)}_t = 0$.
  - $t < W$: the first window's statistics.
  - Pooled intraday model only (rows are intervals (t', i')): median and IQR over rows with $a_{t',i'} = 1$; fewer than 512 such rows give $z^{(k)} = 0$.

- **Column screening:** at every estimation, on the window $\mathcal{W}_t$, keep

$$\mathcal{J}_t = \{j :\ \boldsymbol{x}_{\cdot j} \text{ not constant},\ \boldsymbol{x}_{\cdot j} \neq \boldsymbol{x}_{\cdot j'}\ \forall\, j' < j,\ j' \in \mathcal{J}_t\}, \qquad \boldsymbol{x}_{\cdot j} = (x_{t'j})_{t'\in\mathcal{W}_t}.$$

  - Equality is exact (bitwise); the first copy is kept.
  - The closing-interval model with the full predictor set keeps a median of 369 of its 628 columns.

- **Predictor sets** (Table 3)**:** the HAR benchmark has 22 columns; the full predictor set has 628.

$$628 = \underbrace{22}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{47\times 6}_{\bar{u}^{(k)}} + \underbrace{47\times 6}_{\bar{a}^{(k)}} + \underbrace{7\times 6}_{\bar{\chi}^{(k)}}, \qquad 22 = 6 + 16.$$

Table 3: Composition of the predictor sets.

		Closing-interval model				Pooled intraday
Predictor group	Series	Series averages $u^{(k)}$	Missing-data indicators $\bar{a}^{(k)}$	Activity indicators $\bar{\chi}^{(k)}$	Total	Total
HAR components $\bar{y}^{(k)}$	1	6	–	–	6	12
HAR components × open, close	–	–	–	–	–	24
Calendar dummies $\boldsymbol{D}$	–	–	–	–	16	16
HAR benchmark	–	–	–	–	22	52
Realized moments	7	42	42	–	84	168
Liquidity	14	84	84	6	174	348
Equal-weighted market	6	36	36	–	72	144
Value-weighted market	6	36	36	–	72	144
Sentiment	3	18	18	–	36	72
Implied volatility	3	18	18	–	36	72
Volatility demand	4	24	24	24	72	144
FOMC	4	24	24	12	60	–
Additional predictors	47	282	282	42	606	1,092
Full predictor set	–	–	–	–	628	1,144

Notes: $u^{(k)}$: series u averaged over the previous k intervals; $\bar{a}^{(k)}(u)$ and $\bar{\chi}^{(k)}(u)$: the same averages of its availability (missing-data) and activity indicators (Section 4 and Table 30). Horizons: $k \in \{1, 5, 25, 125, 625, 3125\}$ in the closing-interval model and $k \in \{1, 2, 4, \ldots, 2048\}$ in the pooled intraday model; activity indicators exist for the 7 series whose mode accounts for at least 20% of intervals. In the closing-interval model, the twelve interactions of the HAR components with the opening- and closing-interval dummies are zero or copies of $\bar{y}^{(k)}$ and are omitted, and four of the 16 calendar dummies (hour, overnight, open, close) are constant and are dropped at every estimation. The pooled intraday panel predates the FOMC group (4 series) and has no FOMC predictors; with them it would have 1,264 columns. At every estimation, predictors constant within the rolling window or exact copies of another are dropped.

  - The pooled intraday model's full predictor set has 1,144 columns: 43 series, 12 horizons, and 24 interactions of the HAR components with the opening- and closing-interval dummies.

$$1{,}144 = \underbrace{12 + 16}_{\bar{y}^{(k)},\ \boldsymbol{D}} + \underbrace{24}_{\bar{y}^{(k)} \times \text{open, close}} + 43 \times 12 + 43 \times 12 + 5 \times 12.$$

- **No look-ahead:** every statistic dated (t, i) uses only intervals before (t, i), so every closing-interval predictor is known at the 15:30 ET forecast origin: the trailing averages, $s^2_{t,i}$, the winsorization bounds, $\rho_{t,i}$, the standardization statistics and $\mathcal{J}_t$.
  - Full-sample elements, disclosed: the imputation median, the mode, $\bar{\sigma}^{(u)}$ and the assignment of series to the inverse hyperbolic sine.
  - The first window's standardization statistics use warm-up rows only.

5 Methods

5.1 Problem formulation

- **Rolling-window out-of-sample exercise,** re-estimated daily; every predictor is in the information set $\mathcal{F}_t$ at the 15:30 ET forecast origin.
- **Notation:** $t = 1, \ldots, T$ indexes trading days and $i = 1, \ldots, M$ the 30-minute intraday intervals of day t, overnight included; $i = c$ is the closing interval, the closing half hour (15:30–16:00 ET).
  - $\mathrm{RV}_{t,i} = \sum_{k=1}^{N_{t,i}} r_{t,i,k}^2$: realized variance of interval (t, i), with $r_{t,i,k}$ its kth one-minute log return, $N_{t,i} \leq 30$ (Andersen et al., 2003).
  - Only the closing interval is modeled in the closing-interval model, so $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$.
  - $(t, i-j)$: the jth retained interval before (t, i), possibly on an earlier trading day.
  - $\mathcal{F}_t$: all data through 15:30 ET on day t; $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ and $\mathrm{Var}_t(\cdot) \equiv \mathrm{Var}(\cdot \mid \mathcal{F}_t)$.
- **Intraday periodicity factor** (Andersen and Bollerslev, 1997): the trailing mean of RV in the same interval, strictly before day t,

$$s_{t,i}^2 = \frac{1}{|\mathcal{D}_{t,i}|} \sum_{t' \in \mathcal{D}_{t,i}} \mathrm{RV}_{t',i},$$

$$\mathcal{D}_{t,i} = \{\text{last 20 trading days } t' < t \text{ with interval } i\}, \qquad |\mathcal{D}_{t,i}| \geq 5.$$

  - Intervals with $|\mathcal{D}_{t,i}| < 5$: not scored.
- **Target:** periodicity-adjusted realized volatility (the square root stabilizes the variance), winsorized on that scale at the rolling 1st/99th percentiles:

$$y_{t,i} = \sqrt{\mathrm{RV}_{t,i}/s_{t,i}^2}, \qquad \mathcal{Y}_{t,i} = \{y_{t,i-j} : j = 1, \ldots, 240\},$$
$$\tilde{y}_{t,i} = \mathrm{clip}(y_{t,i},\ Q_{0.01}(\mathcal{Y}_{t,i}),\ Q_{0.99}(\mathcal{Y}_{t,i})).$$

  - $Q_{0.01}(\cdot)$, $Q_{0.99}(\cdot)$: empirical 1st and 99th percentiles; clip truncates its first argument to the interval between the other two.
  - Bounds: rolling quantiles of the previous 240 intervals, all intervals of the day pooled.
  - Models fit $\tilde{y}_{t,i}$; out-of-sample scores use the unwinsorized $\mathrm{RV}_{t,c}$.
  - Closing-interval model: one observation per trading day t, the closing interval; $y_t \equiv y_{t,c}$ and $\tilde{y}_t \equiv \tilde{y}_{t,c}$.
- **Back-transform:** square and rescale; no smearing (Duan, 1983), no recalibration:

$$\hat{\sigma}_t^2 = \hat{y}_t^{\,2}\, s_{t,c}^2,$$

  where $\hat{y}_t$ is the forecast of y_t made with $\mathcal{F}_t$.
  - Jensen bias, downward, not corrected and disclosed: $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t[y_t])^2 = \mathrm{Var}_t(y_t) \geq 0$.
- **Scoring loss:** QLIKE, robust to noise in the realized-variance proxy (Patton, 2011), evaluated at the actual (unwinsorized) realized variance:

$$\mathrm{QLIKE}(\mathrm{RV}, \hat{\sigma}^2) = \frac{\mathrm{RV}}{\hat{\sigma}^2} - \log \frac{\mathrm{RV}}{\hat{\sigma}^2} - 1, \qquad L_t = \mathrm{QLIKE}(\mathrm{RV}_t,\, \hat{\sigma}_t^2) = \mathrm{QLIKE}(y_t^2,\, \hat{y}_t^2).$$

  - $s_{t,c}^2$ cancels, since $\mathrm{RV}_t = y_t^2\, s_{t,c}^2$ with the unwinsorized y_t: the loss is scale-invariant.
  - Scores use RV_t, not the winsorized $\tilde{y}_t^2 s_{t,c}^2$; the models still fit $\tilde{y}$.

- **Rolling estimation window** $\mathcal{W}_t = \{t - W, \ldots, t - 1\}$ of length W.
  - Closing-interval model ($\mathcal{I} = \{c\}$, the intervals used in estimation): the closing interval only; $W = 2{,}000$ trading days; re-estimated daily.
  - Pooled intraday model ($\mathcal{I} = \{1, \ldots, M\}$): all $M = 48$ intervals; $W = 24{,}000$ intervals, $\mathcal{W}_{t,i} = \{(t, i - W), \ldots, (t, i - 1)\}$; re-estimated every interval (trees) or every 21 trading days (network).
- **Hyperparameter-selection splits run forward in time** (Gu et al., 2020): an estimation sample, a 25-observation gap and a 125-observation validation sample:

$$\mathcal{W}_t^{\text{est}} = \{t - W, \ldots, t - n_{\text{val}} - n_{\text{gap}} - 1\},$$
$$\mathcal{W}_t^{\text{gap}} = \{t - n_{\text{val}} - n_{\text{gap}}, \ldots, t - n_{\text{val}} - 1\},$$
$$\mathcal{W}_t^{\text{val}} = \{t - n_{\text{val}}, \ldots, t - 1\},$$

  $n_{\text{gap}} = 25$, $n_{\text{val}} = 125$. Observation t never enters.
- **Estimation objective:** squared error on the scale of $\tilde{y}$. The forecast at origin t is estimated on the full window:

$$\hat{\boldsymbol{\theta}}_t = \arg\min_{\boldsymbol{\theta}} \sum_{t' \in \mathcal{W}_t} \left(\tilde{y}_{t'} - \hat{y}_{t'}(\boldsymbol{\theta})\right)^2.$$

  - t' indexes the observations of a window; $\boldsymbol{\theta}$: the model's parameters.
  - During hyperparameter selection only, each candidate configuration is estimated on the estimation sample $\mathcal{W}_t^{\text{est}}$ in place of $\mathcal{W}_t$ and evaluated on the validation sample $\mathcal{W}_t^{\text{val}}$.
  - Exception: the QLIKE-trained network.

5.2 Models

- **HAR and the HAR benchmark, by OLS** (Corsi, 2009):

$$\hat{y}_t = \beta_0 + \sum_{k \in \mathcal{K}} \beta^{(k)}\, \bar{y}_t^{(k)} + \boldsymbol{\delta}^\top \boldsymbol{D}_t, \qquad \bar{y}_t^{(k)} = \frac{1}{k} \sum_{j=1}^{k} \tilde{y}_{t,c-j}, \qquad \mathcal{K} = \{1, 5, 25, 125, 625, 3125\}.$$

  - $\bar{y}_t^{(k)}$: the HAR component at horizon k, the mean over the k half-hour intervals before the closing interval, overnight included.
  - $\boldsymbol{D}_t$: calendar dummies, coefficients $\boldsymbol{\delta}$; HAR alone sets $\boldsymbol{\delta} = \mathbf{0}$. The HAR benchmark ($\mathcal{X} = \text{HAR}$) is the HAR components and the calendar dummies.
  - $\boldsymbol{x}_t \in \mathbb{R}^p$: the full predictor set ($\mathcal{X} = \text{Full}$), the HAR benchmark's predictors plus, for each additional series u, the HAR components $\bar{u}_t^{(k)}$, $k \in \mathcal{K}$, and the averages of its availability and activity indicators (Section 4).
- **Ridge (Hoerl and Kennard, 1970), lasso (Tibshirani, 1996) and elastic net (Zou and Hastie, 2005),** written in the scikit-learn parametrization (the λ grids below are on this scale) and solved exactly as the window rolls:

$$\hat{\boldsymbol{\beta}}_t^{\text{ridge}} = \arg\min_{\beta_0, \boldsymbol{\beta}} \sum_{t' \in \mathcal{W}_t} (\tilde{y}_{t'} - \beta_0 - \boldsymbol{x}_{t'}^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

$$\hat{\boldsymbol{\beta}}_t^{\text{enet}} = \arg\min_{\beta_0, \boldsymbol{\beta}} \frac{1}{2|\mathcal{W}_t|} \sum_{t' \in \mathcal{W}_t} (\tilde{y}_{t'} - \beta_0 - \boldsymbol{x}_{t'}^\top \boldsymbol{\beta})^2 + \lambda\alpha \sum_{j=1}^{p} \omega_j |\beta_j| + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^{p} \omega_j\, \beta_j^2,$$

  with $\lambda > 0$ the penalty strength, $\alpha \in [0, 1]$ the elastic-net mixing weight, and $\omega_j \geq 0$ the penalty weight on predictor j (intercept β_0 unpenalized, $\omega_0 = 0$):

  - **Block penalty** (block-penalized ridge, lasso and elastic net): the HAR benchmark is penalized 100 times more lightly than the additional predictors,

$$\omega_j = \begin{cases} \underline{\omega} = 0.01 & j \in \text{HAR benchmark}, \\ 1 & j \text{ an additional predictor}, \\ 0 & j = 0 \text{ (intercept)}. \end{cases}$$

  - HAR benchmark alone: $\omega_j = 1$ for every predictor, a single penalty.
  - Lasso: $\alpha = 1$. Elastic net: $\alpha = 0.5$.
  - Ridge: rank-one add and drop on $(\boldsymbol{X}_t^\top \boldsymbol{X}_t + \lambda \operatorname{diag}(\boldsymbol{\omega}))^{-1}$, where $\boldsymbol{X}_t$ stacks the rows $(1, \boldsymbol{x}_{t'}^\top)$, $t' \in \mathcal{W}_t$, and $\boldsymbol{\omega} = (\omega_0, \ldots, \omega_p)^\top$.
  - Lasso, elastic net: exact online homotopy (Garrigues and El Ghaoui, 2008), one observation in, one out.
  - At every fit, only the screened columns $\mathcal{J}_t$ enter: predictors constant within the window or exact copies of another column are dropped; in the closing-interval model this always removes five of the 16 calendar dummies (Section 4).

- **Closing-interval trees:** LightGBM, XGBoost and random forest, on the same observations and windows as the linear models.
  - **Gradient boosting** (Friedman, 2001), in LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016): loss $\frac{1}{2}(\tilde{y} - F)^2$, rounds $b = 1, \ldots, B$, learning rate η:

$$\begin{aligned} F_b &= F_{b-1} + \eta f_b, \\ f_b &= \arg\min_f \sum_{t' \in \mathcal{W}_t} \left[g_{t'} f(\boldsymbol{x}_{t'}) + \tfrac{1}{2} h_{t'} f(\boldsymbol{x}_{t'})^2 \right] \\ &\quad + \gamma^{\text{tree}} J + \lambda_1^{\text{tree}} \|\boldsymbol{w}\|_1 + \tfrac{1}{2} \lambda_2^{\text{tree}} \|\boldsymbol{w}\|_2^2. \end{aligned}$$

  - $g_{t'} = F_{b-1}(\boldsymbol{x}_{t'}) - \tilde{y}_{t'}$ and $h_{t'} = 1$: first and second derivatives of the loss at F_{b-1}; f: a tree with J leaves and leaf weights $\boldsymbol{w}$; γ^{tree}: penalty per leaf; $\lambda_1^{\text{tree}}, \lambda_2^{\text{tree}}$: L1 and L2 penalties on $\boldsymbol{w}$.
  - Leaf weight at $\lambda_1^{\text{tree}} = 0$: $w_j^* = -G_j/(H_j + \lambda_2^{\text{tree}})$, with $G_j = \sum_{t' \in \mathcal{R}_j} g_{t'}$ and $H_j = \sum_{t' \in \mathcal{R}_j} h_{t'} = n_j$ over the n_j observations $\mathcal{R}_j$ in leaf j.
  - Leaf minimum $n_{\min}$: $n_j \geq n_{\min}$ (LightGBM's minimum leaf size); $H_j \geq n_{\min}$ (XGBoost's minimum child weight, an observation count since $h_{t'} = 1$).
  - **Random forest** (Breiman, 2001): average of B trees f_b, each grown on a bootstrap sample,

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^{B} f_b(\boldsymbol{x}_t);$$

    a random subset of the predictors is tried at each split.

- **Pooled intraday trees:** one fixed hyperparameter configuration per tree method, never re-selected, re-estimated every interval on a panel of 1,144 predictors (Table 6).
- **MinGRU network** (the minGRU of Feng et al. (2024), in PufferLib 5.0 (Suarez, 2024), native CUDA): an encoder, one MinGRU layer with highway gates and a decoder, trained with Muon (Jordan et al., 2024). On the predictors of the last ℓ observations, $\boldsymbol{x}_{t-\ell+1:t}$, with

$\tau = t-\ell+1, \dots, t$ and $\boldsymbol{W}_l = [\boldsymbol{W}_l^h;\, \boldsymbol{W}_l^z;\, \boldsymbol{W}_l^\psi]$ split into three row blocks:

$$\begin{aligned}
\boldsymbol{\xi}_\tau &= \boldsymbol{W}_e\,(\boldsymbol{x}_\tau, 1), & \tilde{\boldsymbol{h}}_\tau &= \Lambda^+(\boldsymbol{W}_l^h\,\boldsymbol{\xi}_\tau),\\
\boldsymbol{z}_\tau &= \Lambda(\boldsymbol{W}_l^z\,\boldsymbol{\xi}_\tau), & \boldsymbol{\psi}_\tau &= \Lambda(\boldsymbol{W}_l^\psi\,\boldsymbol{\xi}_\tau),\\
\boldsymbol{h}_\tau &= (1-\boldsymbol{z}_\tau)\odot \boldsymbol{h}_{\tau-1} + \boldsymbol{z}_\tau \odot \tilde{\boldsymbol{h}}_\tau, & \boldsymbol{h}_{t-\ell} &= \boldsymbol{0},\\
\boldsymbol{h}_\tau^* &= \boldsymbol{\psi}_\tau \odot \boldsymbol{h}_\tau + (1-\boldsymbol{\psi}_\tau)\odot \boldsymbol{\xi}_\tau, & o_t &= \boldsymbol{w}_d^\top \boldsymbol{h}_t^* = f^{\mathrm{NN}}(\boldsymbol{x}_{t-\ell+1:t}).
\end{aligned}$$

- $\Lambda(a) = 1/(1+e^{-a})$: the logistic function; $\Lambda^+(a) = a + \frac{1}{2}$ for $a \geq 0$ and $\Lambda(a)$ otherwise, elementwise; $\boldsymbol{z}_\tau$: update gate; $\boldsymbol{\psi}_\tau$: highway gate; $\boldsymbol{h}_\tau \in \mathbb{R}^H$: hidden state of width H; $\boldsymbol{h}_\tau^*$: highway output; $\odot$: elementwise product.
- **Softplus output,** so every forecast is positive ($\mathrm{softplus}(a) = \log(1+e^a)$):

$$\hat{y}_t = \mathrm{softplus}(\varpi_0 + \varpi_1 o_t), \qquad \varpi_0 = \mathrm{softplus}^{-1}(\mu_y), \qquad \varpi_1 = \sigma_y/\Lambda(\varpi_0).$$

  μ_y, σ_y: mean and standard deviation of $\tilde{y}$ over the training observations; $\hat{y}_t \approx \mu_y + \sigma_y o_t$ near $o_t = 0$.
- **Linear shortcut:** a linear term beside the recurrent branch f^{NN},

$$\hat{y}_t = \mathrm{softplus}\big(\varpi_0 + \varpi_1\,[\,\boldsymbol{v}^\top(\boldsymbol{x}_t, 1) + f^{\mathrm{NN}}(\boldsymbol{x}_{t-\ell+1:t})\,]\big).$$

  Zero-initialized decoder: $f^{\mathrm{NN}} \equiv 0$ at the start.
- **Training loss,** per observation t': squared error $\mathcal{L}_{t'} = (\hat{y}_{t'} - \tilde{y}_{t'})^2$, or QLIKE with a floor ε on both the realization and the forecast and a clamp on the log ratio:

$$\mathcal{L}_{t'} = r_{t'} - \log r_{t'} - 1, \qquad \log r_{t'} = \mathrm{clip}\Big(\log \frac{\max(\tilde{y}_{t'}^2,\, \varepsilon^2)}{\hat{y}_{t'}^2 + \varepsilon^2},\ -80,\ 80\Big),$$

  that is, $\mathcal{L}_{t'} = \mathrm{QLIKE}(\max(\tilde{y}_{t'}^2, \varepsilon^2),\ \hat{y}_{t'}^2 + \varepsilon^2)$ whenever the clamp does not bind. $|\partial\mathcal{L}_{t'}/\partial\hat{y}_{t'}|$ capped at a training-window quantile; $\bar{\mathcal{L}}$: the minibatch mean of $\mathcal{L}_{t'}$.
- **Cosine schedule** over $E_{\max} = 200$ epochs, epoch learning rate η_e from η_0:

$$\eta_e = \tfrac{1}{2}\,\eta_0\big(1 + \cos(\pi e/E_{\max})\big), \qquad e = 0, \dots, E_{\max} - 1.$$

- **Muon with decoupled weight decay** ν, every matrix $\boldsymbol{W} \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$:

$$\boldsymbol{M} \leftarrow 0.95\,\boldsymbol{M} + \boldsymbol{G}, \qquad \boldsymbol{W} \leftarrow \boldsymbol{W}(1-\eta_e\,\nu) - \eta_e\sqrt{\max(1, d_{\mathrm{out}}/d_{\mathrm{in}})}\ \mathrm{NS}_5(\boldsymbol{G} + 0.95\,\boldsymbol{M}).$$

  $\boldsymbol{G}$: gradient of $\bar{\mathcal{L}}$, global norm clipped at 1.0; $\boldsymbol{M}$: momentum; NS_5: five Newton–Schulz steps.
- **Encoder input decay** ν_{enc}, column by column, in proportion to the linear models' penalty weights ω_j:

$$\boldsymbol{w}_{e,j} \leftarrow \boldsymbol{w}_{e,j}\,(1 - \eta_e\,\nu_{\mathrm{enc}}\,\omega_j), \qquad j = 1, \dots, p,$$

  with $\boldsymbol{w}_{e,j}$ the jth column of $\boldsymbol{W}_e$.
- **Shortcut penalty** ν_{lin}: L2 under Adam, not decoupled; $\boldsymbol{\omega}$ is 0 on the constant, so the intercept is unpenalized:

$$\nabla_{\boldsymbol{v}} \leftarrow \nabla_{\boldsymbol{v}}\bar{\mathcal{L}} + \nu_{\mathrm{lin}}\,\boldsymbol{\omega} \odot \boldsymbol{v} = \nabla_{\boldsymbol{v}}\Big(\bar{\mathcal{L}} + \tfrac{1}{2}\,\nu_{\mathrm{lin}}\textstyle\sum_j \omega_j v_j^2\Big).$$

  Adam with moment decay rates 0.9 and 0.999, its own gradient clip and its own learning rate η_{lin} on the same cosine schedule.

– **Ridge initialization,** each window's closed form on the window's rows with a full look-back of ℓ observations, $\mathcal{W}_t^\ell = \{t' \in \mathcal{W}_t : t' - \ell + 1 \geq t - W\}$, the first $\ell - 1$ rows dropped (during hyperparameter selection, the rows of $\mathcal{W}_t^{\text{est}}$ with a full look-back):

$$\boldsymbol{v}^{\text{init}} = \arg\min_{\boldsymbol{v}} \frac{1}{|\mathcal{W}_t^\ell|} \sum_{t' \in \mathcal{W}_t^\ell} \left(\boldsymbol{v}^\top (\boldsymbol{x}_{t'}, 1) - \zeta_{t'}\right)^2 + \frac{\lambda^{\text{init}}}{2} \sum_j \omega_j \, v_j^2,$$

$$\zeta_{t'} = \frac{\tilde{y}_{t'} - \mu_y}{\sigma_y}, \qquad \lambda^{\text{init}} = \frac{\nu_{\text{lin}}}{\sigma_y^2}.$$

– **Trained, early-stopped and selected on QLIKE.**

5.3 Tuned hyperparameters

- **Linear penalties,** chosen by hyperparameter selection on a pre-sample, one grid value per decade (Table 4):

$$\lambda^{\text{ridge}} \in \{10^{-2}, \ldots, 10^3\}, \qquad \lambda^{\text{lasso}}, \lambda^{\text{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  – full predictor set: ridge 10^3, lasso and elastic net 10^{-2} (top edge);
  – HAR benchmark: the lower edge of each grid.
- **Trees,** fixed hyperparameters from a search on a 1998–2002 pre-sample (Table 4; search spaces of the re-tuned specifications in Table 5):
  – LightGBM and XGBoost: the best trial of an Optuna search (Akiba et al., 2019);
  – random forest: trial 255 of 400 ($B = 485$ trees, maximum depth 20, share of predictors tried at each split 0.90, bootstrap sample share 0.41);
  – leaf minimums are rescaled to the 2,000-observation window: $n_{\min} \leftarrow n_{\min} \cdot 2{,}000/24{,}000$.
- **Network grid** Θ, with $\boldsymbol{\vartheta} \in \Theta$ a hyperparameter configuration (Table 5, Panel B):
  – closing-interval model: $(\ell, H, \eta_0) \in \{5, 20\} \times \{16, 64\} \times \{0.003, 0.01, 0.03\}$;
  – crossed with ν_{enc}, ν_{lin} and η_{lin};
  – pooled intraday model: $(\ell, H, \eta_0) \in \{20, 48\} \times \{64, 256\} \times \{0.001, 0.003\}$.

5.4 Tuning procedures

- **Windows, re-estimation and re-selection frequencies and criteria** of every model class: Table 6.
- **Linear:**
  – hyperparameter selection on a pre-sample of four folds of 250 trading days (6 January 2016 to 2 January 2020), on validation MSE:

$$\hat{\lambda} = \arg\min_{\lambda} \frac{1}{4} \sum_{j=1}^{4} \text{MSE}_j^{\text{val}}(\lambda),$$

    with $\text{MSE}_j^{\text{val}}(\lambda)$ the mean squared error of $\tilde{y}$ on the validation sample of fold j;
  – a light re-tune at each re-tuning date (every 250 trading days): the grid neighbors $\mathcal{C} \subseteq \{\lambda^-, \lambda^+\}$ (one decade below and above, where on the grid) challenge the incumbent λ;

  - the best challenger λ' (lowest validation MSE) replaces it only if the Diebold–Mariano statistic DM of Section 5.5, on the paired squared errors, is significantly negative:

$$\mathrm{DM}(\{e^2_{t'}(\lambda') - e^2_{t'}(\lambda)\}_{t'}) < -z_{0.05/|\mathcal{C}|}, \qquad z_{0.05/|\mathcal{C}|} = \Phi^{-1}(1 - 0.05/|\mathcal{C}|);$$

  - $e_{t'}(\lambda) = \tilde{y}_{t'} - \hat{y}_{t'}(\lambda)$, t' over two 125-day validation folds; Φ: the standard normal distribution function.

- **Trees,** in increasing order of tuning effort:
  - fixed hyperparameters, re-estimated every 10 trading days or daily;
  - random search every 250 trading days;
  - Optuna (Akiba et al., 2019): 50 trials, re-tuned every 1, 5, 25 or 250 trading days, best of the first 10, 25 or 50 trials (search spaces in Table 5, Panel A).
- **Network:**
  - seed average over S seeds, $\hat{y}_t^{(j)}$ the forecast of seed j:

$$\hat{y}_t = \frac{1}{S}\sum_{j=1}^{S} \hat{y}_t^{(j)};$$

  - the hyperparameters are re-selected at the start of each of 5 blocks of 250 forecast origins, on the validation sample:

$$\hat{\boldsymbol{\vartheta}}_t = \arg\min_{\boldsymbol{\vartheta}\in\Theta} \frac{1}{|\mathcal{W}_t^{\mathrm{val}}|} \sum_{t'\in\mathcal{W}_t^{\mathrm{val}}} \mathcal{L}^{\mathrm{val}}(\tilde{y}_{t'}, \hat{y}_{t'}(\boldsymbol{\vartheta})), \qquad \mathcal{L}^{\mathrm{val}}(y,\hat{y}) \in \{(y-\hat{y})^2,\ \mathrm{QLIKE}(y^2, \hat{y}^{\,2})\};$$

  - each seed's early-stopping epoch $e_j^* = \arg\min_e \bar{\mathcal{L}}_{j,e}^{\mathrm{val}}$ is kept and the network is re-estimated from scratch at every origin; $\bar{\mathcal{L}}_{j,e}^{\mathrm{val}}$: seed j's mean training loss on $\mathcal{W}_t^{\mathrm{val}}$ after epoch e;
  - the forecast is the mean of $S = 5$ seeds (3 in the pooled intraday model).
- **Every selection uses only data available at the forecast origin;** the out-of-sample period is never used for a choice.

5.5 Statistical tests

- **Diebold–Mariano test** (Diebold and Mariano, 1995) on the daily QLIKE loss differential of two forecasts A and B, with the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997):

$$d_t = L_t^A - L_t^B, \qquad \mathrm{DM} = \sqrt{\frac{|\mathcal{O}| + 1 - 2h + h(h-1)/|\mathcal{O}|}{|\mathcal{O}|}}\ \frac{\bar{d}}{\sqrt{\hat{\Omega}/|\mathcal{O}|}}, \qquad h = 1.$$

  - A, B: forecasts $f_{m,\mathcal{X},\mathcal{I}}$, indexed by the model m, the predictor set $\mathcal{X} \in \{\mathrm{HAR}, \mathrm{Full}\}$ and the intervals used in estimation $\mathcal{I} \in \{\{c\}, \{1,\dots,M\}\}$.
  - $\mathcal{O}$: the out-of-sample days of the exercise, numbered $t = 1,\dots,|\mathcal{O}|$ in time order; $\mathcal{O} = \mathcal{T}_{\mathrm{oos}}$, $|\mathcal{T}_{\mathrm{oos}}| = T_{\mathrm{oos}}$, in the headline comparison.
  - $\bar{d}$: mean loss differential over $\mathcal{O}$; h: forecast horizon; two-sided p-value from the standard normal.

- **Newey–West variance** (Newey and West, 1987), Bartlett weights, lag truncation q_{NW}:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\mathrm{NW}}}\Big(1 - \frac{j}{q_{\mathrm{NW}}+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{|\mathcal{O}|}\sum_{t=j+1}^{|\mathcal{O}|}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

  - DM: $q_{\mathrm{NW}} = \max\big(h-1, \lfloor 4(|\mathcal{O}|/100)^{2/9}\rfloor\big)$, which gives $q_{\mathrm{NW}} = 6$ on $\mathcal{T}_{\mathrm{oos}}$.
- **HAC t-statistic of a mean daily return** (Newey–West, no HLN correction): $t = \bar{r}/\sqrt{\hat{\Omega}_r/|\mathcal{O}|}$, with $\hat{\Omega}_r$ the Newey–West variance above computed on a daily return series r_t in place of d_t: a single strategy's return (a test of its mean return) or the difference between two strategies' returns (a test of the mean return differential); $q_{\mathrm{NW}} = \lfloor 1.5\,|\mathcal{O}|^{1/3}\rfloor$, which gives $q_{\mathrm{NW}} = 14$ on $\mathcal{T}_{\mathrm{oos}}$.
- **Model confidence set** (Hansen et al., 2011), **superior predictive ability** (Hansen, 2005) and **StepM** (Romano and Wolf, 2005) for the comparison of predictor sets.
- **Point estimates only;** no bootstrap confidence intervals in the main tables.

6 Results

6.1 Forecast evaluation

- **Forecast and realization.** The variance forecast is the back-transformed fit, $\hat{\sigma}_t^2 = \hat{y}_t^2\, s_{t,c}^2$, with no bias correction. The realization is the actual realized variance of the closing half hour (15:30–16:00 ET), $\mathrm{RV}_t \equiv \mathrm{RV}_{t,c}$; it is not winsorized (the winsorized $\tilde{y}_{t,c}$ enters estimation only). The out-of-sample period is the set $\mathcal{T}_{\mathrm{oos}}$ of $T_{\mathrm{oos}} = |\mathcal{T}_{\mathrm{oos}}| = 866$ trading days; in the HAC formulas, $t-j$ is the out-of-sample day j days before t.
- **Loss.** The daily QLIKE loss (Patton, 2011) and its out-of-sample average:

$$L_t = \mathrm{QLIKE}(\mathrm{RV}_t, \hat{\sigma}_t^2) = \frac{\mathrm{RV}_t}{\hat{\sigma}_t^2} - \log\frac{\mathrm{RV}_t}{\hat{\sigma}_t^2} - 1, \qquad \overline{\mathrm{QLIKE}} = \bar{L} = \frac{1}{T_{\mathrm{oos}}}\sum_{t\in\mathcal{T}_{\mathrm{oos}}} L_t.$$

- **Benchmark forecast.** The closing-interval OLS regression on the HAR benchmark, the HAR components and the calendar dummies (Corsi, 2009). Its average QLIKE is $\overline{\mathrm{QLIKE}} = 0.0967$, and its sign strategy has an annualized Sharpe ratio at mid-quotes of SR $= 1.37$ (Section 7).
- **Relative loss.** For forecast A against forecast B (the benchmark forecast unless stated otherwise), the loss differential is $d_t = L_t^A - L_t^B$, and the percentage change in average QLIKE is

$$\Delta\% = 100\,\frac{\bar{d}}{\bar{L}^B}, \qquad \bar{d} = \frac{1}{T_{\mathrm{oos}}}\sum_{t\in\mathcal{T}_{\mathrm{oos}}} d_t.$$

- **Diebold–Mariano test** (Diebold and Mariano, 1995), with the Harvey–Leybourne–Newbold small-sample correction at horizon $h = 1$ (Harvey et al., 1997) and a two-sided normal p-value:

$$\mathrm{DM} = \sqrt{\frac{T_{\mathrm{oos}}-1}{T_{\mathrm{oos}}}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/T_{\mathrm{oos}}}}, \qquad p = 2\{1 - \Phi(|\mathrm{DM}|)\}.$$

- **Long-run variance:** Newey and West (1987), with Bartlett weights:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q_{\mathrm{NW}}} w_j\,\hat{\gamma}_j, \qquad w_j = 1 - \frac{j}{q_{\mathrm{NW}}+1}, \qquad \hat{\gamma}_j = \frac{1}{T_{\mathrm{oos}}-j}\sum_{t=j+1}^{T_{\mathrm{oos}}}(d_t - \bar{d})(d_{t-j} - \bar{d}).$$

Table 4: Hyperparameters: penalized regressions and tree ensembles.

Model	Hyperparameter	Search space	Value: HAR benchmark	Value: Full predictor set
Panel A: Penalized linear models, selected on the pre-sample				
OLS	–	–	–	–
Ridge	Penalty, λ	$\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6)	10^{-2}	10^{3}
Lasso	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
Elastic net	Penalty, λ	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	10^{-6}	10^{-2}
	Mixing weight, α	fixed	0.5	0.5
All three	Penalty weight on the HAR benchmark, $\underline{\omega}$	fixed	1	0.01
Panel B: Tree ensembles, fixed hyperparameters (selected on the 1998–2002 pre-sample)				
LightGBM	Leaves per tree, J	[15, 255] (log)	20	
	Learning rate, η	[0.01, 0.3] (log)	0.013	
	Boosting rounds, B	[50, 500]	392	
	Minimum leaf size, $n_{\min}$	[20, 500] (log)	98 (8)	
	Share of predictors per tree	[0.5, 1]	0.70	
	Share of observations per tree	[0.5, 1]	0.56	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	10^{-8}	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	0.016	
XGBoost	Maximum depth	[3, 10]	5	
	Learning rate, η	[0.01, 0.3] (log)	0.0123	
	Boosting rounds, B	[50, 500]	367	
	Minimum leaf size, $n_{\min}$	[1, 100] (log)	50.2 (4.18)	
	Share of observations per tree	[0.5, 1]	0.66	
	Share of predictors per tree	[0.5, 1]	0.65	
	L_1 penalty on leaf weights, λ_1^{tree}	$[10^{-8}, 10]$ (log)	0.0215	
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-8}, 10]$ (log)	1.2×10^{-5}	
	Penalty per leaf, γ	$[10^{-8}, 1]$ (log)	1.5×10^{-6}	
Random forest	Number of trees, N	[50, 500]	485	
	Maximum depth	[2, 32] (log)	20	
	Share of predictors per split	[0.02, 1] (log)	0.90	
	Bootstrap sample share	[0.05, 1] (log)	0.41	
	Minimum leaf size, $n_{\min}$	[1, 1000] (log)	12 (1)	

Notes: Panel A: λ is on the scale of the objectives in Section 5.2, one grid value per decade (grid size in parentheses), selected by mean validation MSE over the 4 folds of the pre-sample (Table 2). At each of the 4 re-selection points a grid neighbor may replace it (Section 5.4); this happened 4 times in 24 re-selections (ridge 1 and 0, lasso 1 and 0, elastic net 2 and 0, for the HAR benchmark and the full predictor set). Panel B: the best trial of an Optuna search on the pooled intraday pre-sample (24,000-interval window; random forest: trial 255 of 400), ranked by MSE and then held fixed for both predictor sets; (log): searched on a logarithmic scale. $n_{\min}$ was selected on 24,000 intervals and is rescaled to the 2,000-day window (rescaled value in parentheses). Implementations: LightGBM (Ke et al., 2017) with bagging at every round, XGBoost (Chen and Guestrin, 2016) with histogram-based split finding and the scikit-learn random forest with bootstrap samples; random seeds are fixed.

Table 5: Hyperparameter search in the rolling-window exercise.

Model	Hyperparameter	Search space	Selected (count)
Panel A: Tree ensembles, closing-interval models (50 Optuna trials per re-selection)			
LightGBM	Leaves per tree, J	[2, 128] (log)	–
	Minimum leaf size, $n_{\min}$	[1, 256] (log)	–
	Share of predictors per tree	[0.05, 1]	–
	Share of observations per tree	[0.3, 1]	–
	L_2 penalty on leaf weights, λ_2^{tree}	$[0.001, 10^4]$ (log)	–
	Learning rate, η	[0.005, 0.1] (log)	–
XGBoost	Maximum depth	[1, 8]	–
	Minimum leaf size, $n_{\min}$	[0.1, 256] (log)	–
	Share of observations per tree	[0.15, 1]	–
	Share of predictors per tree	[0.05, 1]	–
	L_2 penalty on leaf weights, λ_2^{tree}	$[10^{-5}, 10^4]$ (log)	–
	Learning rate, η	[0.005, 0.1] (log)	–
Random forest	Share of predictors per split	[0.02, 1]	–
	Minimum leaf size, $n_{\min}$	[1, 256] (log)	–
	Maximum depth	$\{3, 5, 10, 20, 40, \text{unrestricted}\}$	–
Random search	The same hyperparameters	32 candidates	–
Panel B: MinGRU network (ridge-initialized, trained and selected on QLIKE)			
Closing-interval model (216 configurations)			
	Look-back length, ℓ	$\{5, 20\}$	20 (8/10)
	Hidden width, H	$\{16, 64\}$	64 (7/10)
	Learning rate, η_0	$\{0.003, 0.01, 0.03\}$	0.03 (9/10)
	Encoder weight decay, ν_{enc}	$\{0, 3, 15\}$	0 (4/10)
	Shortcut weight decay, ν_{lin}	$\{0.02, 0.2, 2\}$	2 (5/10)
	Shortcut learning rate, η_{lin}	$\{0.0003, 0.001\}$	0.0003 (5/10)
Pooled intraday model (144 configurations)			
	Look-back length, ℓ	$\{20, 48\}$	20 (7/10)
	Hidden width, H	$\{64, 256\}$	256 (6/10)
	Learning rate, η_0	$\{0.001, 0.003\}$	0.003 (10/10)
	Encoder weight decay, ν_{enc}	$\{0, 3, 15\}$	3 (5/10)
	Shortcut weight decay, ν_{lin}	$\{0.03, 0.3, 3\}$	3 (6/10)
	Shortcut learning rate, η_{lin}	$\{0, 10^{-5}\}$	10^{-5} (7/10)

Notes: Panel A: at each re-selection, 50 trials of Optuna's TPE sampler start from the fixed hyperparameters of Table 4 and are ranked by validation MSE (in one specification, QLIKE); hyperparameters not listed stay fixed, and B is set by early stopping with patience $\lceil 1/\eta \rceil$ and capped at $\lceil 4B^{\text{fix}}\eta^{\text{fix}}/\eta \rceil$; (log): log-uniform. Random search: the fixed configuration and 31 candidates drawn once from product grids of 700 (LightGBM), 2,800 (XGBoost) and 112 (random forest) configurations. Panel B: the network of Section 5.2; fixed: 1 MinGRU layer, batch size 128, at most $E_{\max} = 200$ epochs (pooled intraday model: 100) with patience 20 (10), Muon momentum 0.95, gradient-norm clip 1.0, cosine learning-rate schedule. Selected: the modal choice over the 5 blocks and 2 predictor sets, with its count.

Table 6: Hyperparameter tuning, by model class.

Model	W	Re-estimated every	Re-selected every	Validation sample / gap	Criterion	S
Panel A: Closing-interval models (trading days)						
OLS	2,000	1	–	–	–	–
Ridge, Lasso, Elastic net	2,000	1	once (pre-sample); then 250	4 × 250 / 25; then 2 × 125 / 25	MSE; switch if DM $< -z_{0.05/\lvert\mathcal{C}\rvert}$	–
Trees, fixed hyperparameters	2,000	10 or 1	–	–	–	1
Trees, random search	2,000	10	250	125 / 25	MSE	1
Trees, Optuna	2,000	1	1, 5, 25, 250	125 / 25	MSE (one specification: QLIKE)	1
MinGRU	2,000	1	250 (5 blocks)	125 / 25	QLIKE of the seed mean	5
Panel B: Pooled intraday models (30-minute intervals)						
Ridge, Lasso, Elastic net	96,000	1	once (pre-sample); then 250	4 × 12,000 / 1,200; then 125 / 25	as in Panel A	–
Trees, fixed hyperparameters (one configuration per method)	24,000	1	–	–	–	1
MinGRU	24,000	21 days	12 re-estimations (5 blocks)	2,400 / 48	QLIKE of the seed mean	3

Notes: Every split is forward: the estimation sample, a gap and the validation sample at the end of the rolling window $\mathcal{W}_t$ (Section 5.4), so the out-of-sample period is never used for a choice. Re-estimated and re-selected every: the spacing of successive estimations and hyperparameter selections, in the units of the panel; $\mathcal{C}$: the grid neighbors that challenge the selected penalty. Criteria are computed on the scale of $\tilde{y}_{t,i}$ unless QLIKE is named; S: the number of random seeds averaged; all random seeds are fixed. Optuna starts a new search at each re-selection; the MinGRU network is re-estimated from scratch at each origin, each seed at its early-stopping epoch; the pooled trees use one fixed configuration per method (for LightGBM and XGBoost, single arms of a 20-configuration development bank), re-estimated at every interval (part of them only every 10th).

- **Lag truncation** $q_{\mathrm{NW}} = \max(h-1, \lfloor 4(T_{\mathrm{oos}}/100)^{2/9} \rfloor) = 6$. DM < 0 favors forecast A. "Significant" means significant at the 5% level ($p < 0.05$).
- **Squared-error and absolute-error loss,** in variance units:

$$\mathrm{MSE} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} (\mathrm{RV}_t - \hat{\sigma}_t^2)^2, \qquad \mathrm{MAE} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} |\mathrm{RV}_t - \hat{\sigma}_t^2|.$$

- **Out-of-sample R^2,** with the out-of-sample mean of the realized variance as the benchmark forecast:

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\mathrm{oos}}} (\mathrm{RV}_t - \hat{\sigma}_t^2)^2}{\sum_{t \in \mathcal{T}_{\mathrm{oos}}} (\mathrm{RV}_t - \overline{\mathrm{RV}})^2}, \qquad \overline{\mathrm{RV}} = \frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \mathrm{RV}_t.$$

- **Calibration:** the ratio of average realized to average forecast variance, $\overline{\mathrm{RV}}/\overline{\hat{\sigma}^2}$, and the frequency of under-prediction, $\frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \mathbf{1}\{\mathrm{RV}_t > \hat{\sigma}_t^2\}$.

6.2 The full predictor set versus the HAR benchmark

- **Paired comparison** within each model class: the same model m and estimation sample $\mathcal{I}$ on the two predictor sets, $d_t = L_t^{\mathrm{Full}} - L_t^{\mathrm{HAR}}$, with $L_t^{\mathcal{X}}$ the loss of the forecast $f_{m,\mathcal{X},\mathcal{I}}$ (Table 7):

$$\Delta\% = 100\Big(\frac{\bar{L}^{\mathrm{Full}}}{\bar{L}^{\mathrm{HAR}}} - 1\Big) = 100\,\frac{\bar{d}}{\bar{L}^{\mathrm{HAR}}}, \qquad \text{DM on } d_t,\ q_{\mathrm{NW}} = 6.$$

- **The full predictor set lowers average QLIKE in most model classes** ($\Delta\% < 0$). The reductions are largest for the pooled intraday trees, $\Delta\% = -14.6$, -12.7 and -13.0 for LightGBM, XGBoost and the random forest (DM $= -4.64$, -4.55, -4.63), and only these are significant at the 5% level.
- **Across all 51 estimator pairs** of the model set, the full predictor set lowers average QLIKE in 38 pairs, 3 of them significantly (DM). It raises the annualized Sharpe ratio at mid-quotes in 50 pairs, 4 of them significantly (HAC t-statistic on the daily return differential, Newey–West, $q_{\mathrm{NW}} = 14$).
- **Squared-error loss disagrees:** the full predictor set raises the MSE in 51 of the 51 pairs, 0 of them significantly.
  - Five days in March 2020 carry 85–92% of $\sum_{t \in \mathcal{T}_{\mathrm{oos}}} (\mathrm{RV}_t - \hat{\sigma}_t^2)^2$.
  - Excluding February–April 2020, the lasso on the full predictor set has a lower MSE than the benchmark forecast.
- **MinGRU network** (ridge-initialized linear shortcut, trained and selected on QLIKE): the full predictor set changes its average QLIKE by -4.0% (DM $= -0.99$, $p = 0.32$), a reduction that is not significant at the 5% level. On the full predictor set, its average QLIKE (0.0966) lies between those of the elastic net (0.0958) and ridge (0.0975).

6.3 Closing-interval versus pooled intraday models

- **The pooled intraday model has a higher average QLIKE than the closing-interval model in every model class,** $\Delta\% = 100(\bar{L}^{\mathrm{pool}}/\bar{L}^{\mathrm{close}} - 1) > 0$, with $\bar{L}^{\mathrm{pool}}$ and $\bar{L}^{\mathrm{close}}$ the average losses of $f_{m,\mathcal{X},\{1,\ldots,M\}}$ and $f_{m,\mathcal{X},\{c\}}$ (Table 8):
  - linear models: +13.0 to +13.6% on the HAR-benchmark predictors (DM from 3.37 to 3.45, significant at the 5% level), +7.2 to +14.9% on the full predictor set (significant for the lasso and the elastic net);

  - trees: +25.0 to +27.9% on the HAR-benchmark predictors and +11.3 to +14.1% on the full predictor set, each significant at the 5% level.
- **Economic value:** for the linear models and the trees, none of the differences in the annualized Sharpe ratio between the pooled intraday and the closing-interval model is significant at the 5% level (HAC t-statistic on the daily return differential, $q_{\mathrm{NW}} = 14$).
- **Pooled intraday MinGRU network:** *[pending: the pooled MinGRU network, ridge-initialized and trained and selected on QLIKE].*

6.4 Model classes

- **The most accurate forecasts, all closing-interval models on the full predictor set, are statistically indistinguishable from the benchmark forecast** ($\overline{\mathrm{QLIKE}} = 0.0967$, rank 9 of 120; Table 9):
  - average QLIKE: lasso 0.0931, XGBoost 0.0957, elastic net 0.0958, LightGBM 0.0961, MinGRU network 0.0966;
  - relative to the benchmark forecast, the lasso changes average QLIKE by -3.7% (DM $= -1.35$, $p = 0.18$), and none of the five is significantly more accurate than the benchmark forecast at the 5% level. Of the 111 forecasts of the model set other than the benchmark forecast, 7 have a lower average QLIKE, 0 of them significantly, and 51 are significantly less accurate.
- **Trees versus the closing-interval ridge on the same predictors:** the tree has the higher average QLIKE in 46 of 48 comparisons, 21 of them significantly.
- **Pooled intraday models trail closing-interval models;** the least accurate are the pooled intraday trees on the HAR-benchmark predictors (average QLIKE 0.125–0.128).

7 Economic value of the forecasts

7.1 The straddle strategy

- **Position:** SPXW options that expire the same day (0DTE). With the index at S_t at the 15:30 ET forecast origin, the call and the put at the nearest strikes on either side,

$$K_C = \min\{K \geq S_t\}, \qquad K_P = \max\{K \leq S_t\},$$

  over strikes with live quotes. If $K_C = K_P = S_t$ the package is an at-the-money straddle; otherwise it is a strangle on the two strikes adjacent to S_t. We call both the straddle: a position that is approximately delta-neutral at entry, entered at 15:30 ET and held, not hedged, to expiration.
- **Entry prices:** at the mid-quote, P_t, and at the quoted bid and ask of the package, P_t^b and P_t^a, from the legs' bid quotes b and ask quotes a at 15:30 ET:

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement** in cash at the official close S_t^{close}, with no delta hedging; R_t is the straddle's simple return on the premium paid at the mid-quote:

$$X_t = (S_t^{\mathrm{close}} - K_C)^+ + (K_P - S_t^{\mathrm{close}})^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

Table 7: Forecast accuracy: HAR benchmark versus full predictor set

	QLIKE				Sharpe ratio	
Model	HAR	Full	Δ (%)	DM	ΔSR	t
Panel A: Closing-interval models						
Ridge	0.0967	0.0975	0.8	(0.18)	0.05	(0.09)
Lasso	0.0967	0.0931	−3.7	(−1.35)	0.50	(1.47)
Elastic net	0.0967	0.0958	−0.9	(−0.20)	0.55	(1.34)
LightGBM	0.1010	0.0961	−4.9	(−1.45)	0.54	(1.11)
XGBoost	0.1001	0.0957	−4.3	(−1.44)	0.16	(0.30)
Random forest	0.0992	0.0991	0.0	(−0.01)	0.18	(0.34)
MinGRU	0.1006	0.0966	−4.0	(−0.99)	0.49	(0.95)
Panel B: Pooled intraday models						
Ridge	0.1098	0.1045	−4.8	(−1.45)	0.53	(1.12)
Lasso	0.1092	0.1069	−2.1	(−0.63)	0.51	(1.19)
Elastic net	0.1096	0.1040	−5.1	(−1.57)	0.61	(1.21)
LightGBM[a]	0.1276	0.1090	−14.6***	(−4.64)	0.51	(1.02)
XGBoost[a]	0.1251	0.1092	−12.7***	(−4.55)	0.30	(0.87)
Random forest[†]	0.1269	0.1103	−13.0***	(−4.63)	0.30	(0.70)
MinGRU	0.1177	0.1141	−3.0	(−0.72)	−0.45	(−1.11)

Notes: The out-of-sample period runs from 3 January 2020 to 30 April 2024 ($T_{\text{oos}} = 866$ trading days). QLIKE is the mean loss $\bar{L}$, with $L_t = \text{QLIKE}(\text{RV}_t, \hat{\sigma}_t^2)$, RV_t the actual realized variance of the closing half hour and $\hat{\sigma}_t^2 = \hat{y}_t^2 s_{t,c}^2$ the variance forecast. $\Delta = 100(\bar{L}^{\text{Full}}/\bar{L}^{\text{HAR}} - 1)$, negative when the full predictor set is more accurate, and DM is the Diebold–Mariano statistic of the daily loss differential (Harvey–Leybourne–Newbold correction; Newey–West lag $q_{\text{NW}} = 6$). ΔSR is the change in the annualized Sharpe ratio of the sign strategy at mid-quotes and t the HAC t-statistic of the daily return differential ($q_{\text{NW}} = 14$; Section 5.5). *, ** and *** denote significance at the 10%, 5% and 1% levels. Closing-interval trees use fixed hyperparameters and are re-estimated daily, and the MinGRU is ridge-initialized and trained and selected on QLIKE (Section 5.2).

[a] Full predictor set: the earlier draft's pooled intraday specification.

[†] Preliminary: the closing-interval forecasts of the pooled intraday random forest on the full predictor set, from an exercise still in progress.

Table 8: Forecast accuracy: closing-interval versus pooled intraday models

	QLIKE		Sharpe ratio	
Model	Δ (%)	DM	ΔSR	t
Panel A: HAR benchmark				
Ridge	13.6***	(3.37)	−0.46	(−1.05)
Lasso	13.0***	(3.39)	−0.68*	(−1.69)
Elastic net	13.4***	(3.45)	−0.49	(−1.12)
LightGBM	26.2***	(4.20)	0.12	(0.36)
XGBoost	25.0***	(4.34)	0.01	(0.01)
Random forest	27.9***	(4.56)	0.35	(0.95)
MinGRU	17.0***	(3.13)	−0.26	(−0.62)
Panel B: Full predictor set				
Ridge	7.2	(1.61)	0.03	(0.07)
Lasso	14.9***	(3.09)	−0.68*	(−1.66)
Elastic net	8.5**	(2.03)	−0.42	(−0.88)
LightGBM[a]	13.4***	(2.84)	0.10	(0.15)
XGBoost[a]	14.1***	(3.03)	0.15	(0.30)
Random forest[†]	11.3***	(2.77)	0.46	(0.85)
MinGRU	18.1***	(4.02)	−1.19**	(−2.30)

Notes: Sample, loss and tests as in Table 7. Each row compares the pooled intraday model with the closing-interval model of the same class on the predictor set of the panel: $\Delta = 100(\bar{L}^{\text{pool}}/\bar{L}^{\text{close}} - 1)$, positive when the closing-interval model is more accurate, and ΔSR is the pooled intraday minus the closing-interval Sharpe ratio. *, ** and *** denote significance at the 10%, 5% and 1% levels. Closing-interval trees use fixed hyperparameters and are re-estimated daily; the pooled intraday trees are re-estimated at every interval and the pooled intraday MinGRU monthly, each on a rolling window of past intervals (Section 5.1).

[a] The earlier draft's pooled intraday specification.

[†] Preliminary (see Table 7).

Table 9: Ranking of the main forecasts by mean QLIKE

				vs. HAR benchmark	
Rank	Model	Predictors	QLIKE	Δ (%)	DM
Panel A: Closing-interval models					
1	Lasso	Full	0.0931	−3.7	(−1.35)
2	XGBoost	Full	0.0957	−1.0	(−0.31)
3	Elastic net	Full	0.0958	−0.9	(−0.21)
4	LightGBM	Full	0.0961	−0.6	(−0.16)
5	MinGRU	Full	0.0966	−0.1	(−0.02)
9	OLS (benchmark)	HAR	0.0967	–	–
10	Ridge	Full	0.0975	0.8	(0.18)
16	Random forest	Full	0.0991	2.5	(0.71)
17	Random forest	HAR	0.0992	2.6	(1.23)
22	XGBoost	HAR	0.1001	3.5	(1.53)
30	MinGRU	HAR	0.1006	4.1*	(1.72)
36	LightGBM	HAR	0.1010	4.5**	(2.04)
Panel B: Pooled intraday models					
76	Elastic net	Full	0.1040	7.5	(1.51)
81	Ridge	Full	0.1045	8.1*	(1.69)
92	Lasso	Full	0.1069	10.6*	(1.96)
100	LightGBM[a]	Full	0.1090	12.7**	(2.33)
102	Ridge (block penalty)[a]	Full	0.1092	13.0**	(2.41)
104	Lasso	HAR	0.1092	13.0***	(3.39)
105	XGBoost[a]	Full	0.1092	13.0**	(2.47)
106	Elastic net	HAR	0.1096	13.3***	(3.44)
108	Ridge	HAR	0.1098	13.6***	(3.37)
112	Random forest[†]	Full	0.1103	14.1**	(2.53)
115	MinGRU	Full	0.1141	18.0***	(4.53)
116	MinGRU	HAR	0.1177	21.7***	(4.80)
117	XGBoost	HAR	0.1251	29.4***	(4.90)
118	Random forest	HAR	0.1269	31.2***	(4.86)
119	OLS[a]	HAR	0.1270	31.4***	(4.50)
120	LightGBM	HAR	0.1276	31.9***	(4.85)

Notes: Sample, loss and tests as in Table 7. $\Delta = 100(\bar{L}/\bar{L}^{\mathrm{OLS}} - 1)$ compares each forecast with the HAR benchmark, the closing-interval OLS on the HAR predictor set, with DM in parentheses; negative values favor the forecast. Rank is the position among the 120 forecasts of Table 31; the closing-interval ridge, lasso and elastic net on the HAR predictor set are within 0.03% of the benchmark and are omitted. Closing-interval trees use fixed hyperparameters and are re-estimated daily. *, ** and *** denote significance at the 10%, 5% and 1% levels.

[a] The earlier draft's pooled intraday specification.

[†] Preliminary (see Table 7).

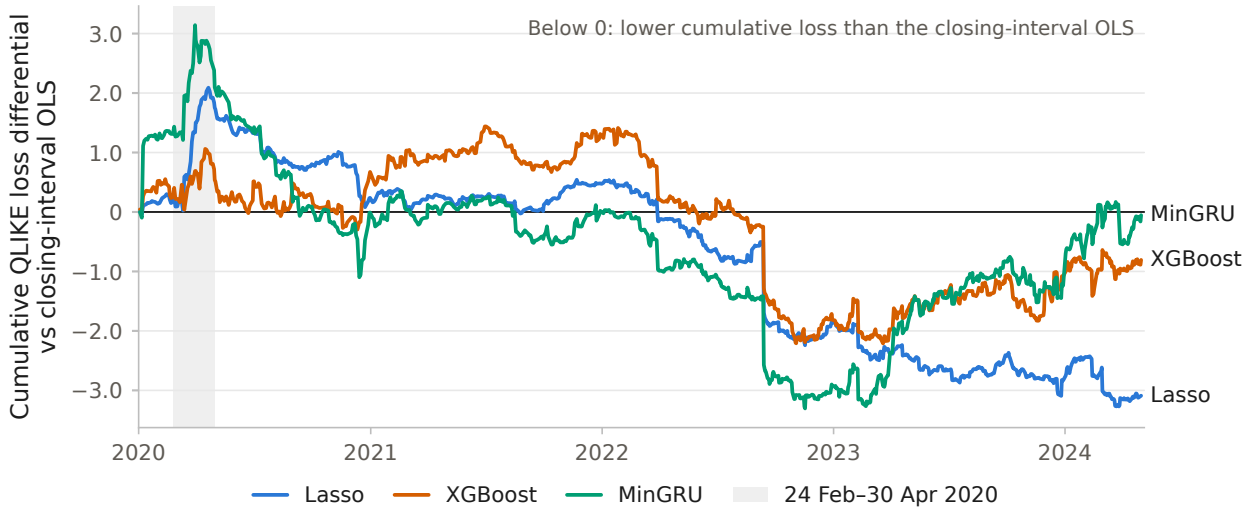


Figure 1: Cumulative QLIKE loss differential relative to the closing-interval OLS on the HAR benchmark. The figure plots $\sum_{t' \le t} d_{t'}$, where $d_{t'} = L_{t'}^{\text{model}} - L_{t'}^{\text{OLS}}$ is the daily loss differential, as in Welch and Goyal (2008), for the closing-interval model with the lowest mean QLIKE in each model class, each on the full predictor set: the lasso, XGBoost (fixed hyperparameters, re-estimated daily) and the MinGRU network (ridge-initialized, trained and selected on QLIKE). Values below zero mean that the model has accumulated a smaller loss than the OLS; the final value divided by 866 equals the difference in mean QLIKE. Losses are computed against actual realized variance. Over the shaded window, 24 February to 30 April 2020, the cumulative differentials change by +1.60 (lasso), +0.30 (XGBoost) and +0.78 (MinGRU); over the full out-of-sample period they end at −3.09, −0.81 and −0.05.

- **Implied variance:** IV_t^C and IV_t^P are the legs' implied volatilities at 15:30 ET, quoted per hour; the implied volatility of the closing half hour is

$$\text{IV}_t = \frac{\text{IV}_t^C + \text{IV}_t^P}{2\sqrt{2}},$$

  and IV_t^2 is the implied variance to expiration.
- **Forecast variance risk premium:** the variance risk premium of the closing half hour is $\text{IV}_t^2 - \mathbb{E}_t[\text{RV}_t]$ (Carr and Wu, 2009; Bollerslev et al., 2009), and its forecast is $\text{IV}_t^2 - \hat{\sigma}_t^2$, with $\hat{\sigma}_t^2$ the variance forecast of Section 5.1. Every strategy below conditions on the forecast only through the gap $\hat{\sigma}_t^2 - \text{IV}_t^2$: it buys when the forecast exceeds implied variance.
- **Strategies:** position q_t in straddles ($q_t > 0$ long, $q_t < 0$ short) and daily return $r_t = q_t R_t$. The forecast enters only through q_t, so the comparison across strategies measures the economic value of the forecast (Fleming et al., 2001):
  - the unconditional short-straddle benchmark, a static short-variance position: $q_t \equiv -1$;
  - sign strategy: $q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2)$, with $q_t = -1$ on ties;
  - robust clip: $q_t = \text{clip}((\hat{\sigma}_t^2 - \text{IV}_t^2)/m_t, -1, 1)$, $m_t = \text{median}_{t'<t}|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2|$;
  - full long, scaled short: $q_t = 1$ if $\hat{\sigma}_t^2 > \text{IV}_t^2$, else $q_t = \max((\hat{\sigma}_t^2 - \text{IV}_t^2)/m_t, -1)$;
  - rank of the absolute gap, over the $t-1$ earlier out-of-sample days $t' < t$:

$$q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2) \cdot \frac{1}{t-1} \sum_{t'<t} \mathbf{1}\{|\hat{\sigma}_{t'}^2 - \text{IV}_{t'}^2| \le |\hat{\sigma}_t^2 - \text{IV}_t^2|\}.$$

- **Sizing sample:** q_t uses days $t' < t$ only. The first 252 out-of-sample days build the scale

m_t, so every sizing rule, the sign strategy included, is evaluated on the remaining 614 trading days, $t > 252$ (Table 13).

- **Execution:** at the mid-quote, $r_t = q_t R_t$. At quoted bid/ask prices a purchase pays the ask and a sale receives the bid; the spread is paid at entry only, since settlement is in cash:

$$r_t^{\mathrm{ba}} = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- **Transaction costs:** trading at the quoted bid or ask instead of the mid-quote costs a median 2.9% of the premium. No other transaction costs are deducted.

7.2 Performance measures

- **Annualized Sharpe ratio,** from daily returns, at mid-quotes and at quoted bid/ask prices, over the out-of-sample days $t \in \mathcal{T}_{\mathrm{oos}}$, $|\mathcal{T}_{\mathrm{oos}}| = T_{\mathrm{oos}}$ (the 614 days with $t > 252$ for the sizing rules):

$$\mathrm{SR} = \sqrt{252}\,\frac{\bar{r}}{\mathrm{sd}(r)}, \qquad \mathrm{sd}(r)^2 = \frac{1}{T_{\mathrm{oos}} - 1} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} (r_t - \bar{r})^2.$$

  The risk-free return over the half-hour holding period is negligible and is not subtracted.
- **Mean return and its HAC t-statistic:** $t = \bar{r}/\sqrt{\hat{\Omega}_r / T_{\mathrm{oos}}}$, with $\hat{\Omega}_r$ the Newey–West long-run variance of Section 5.5 applied to r_t and lag truncation $q_{\mathrm{NW}} = \lfloor 1.5\, T_{\mathrm{oos}}^{1/3} \rfloor = 14$.
- **Hit rate** $\frac{1}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \mathbf{1}\{r_t > 0\}$; number of long positions $\#\{t : q_t > 0\}$.
- **Gain over the short-straddle benchmark** (or over any benchmark strategy B), with the daily return differential $\Delta_t = r_t^A - r_t^B$:

$$\Delta\mathrm{SR} = \mathrm{SR}(r^A) - \mathrm{SR}(r^B), \qquad t_{\mathrm{HAC}} = \frac{\bar{\Delta}}{\sqrt{\hat{\Omega}_\Delta / T_{\mathrm{oos}}}},$$

  the HAC t-statistic on the daily return differential (Newey–West, $q_{\mathrm{NW}} = 14$). It tests the mean return differential, not the equality of the two Sharpe ratios (Ledoit and Wolf, 2008).
- **Elementary score** of the Murphy diagram (Ehm et al., 2016) at threshold θ on the scaled values, evaluated at the implied variance IV_t^2, that is, at $\theta = 1$:

$$\mathrm{ES}_{\theta,t} = |\varrho_t - \theta|\ \mathbf{1}\{\min(\hat{\varrho}_t, \varrho_t) \le \theta < \max(\hat{\varrho}_t, \varrho_t)\}, \qquad \hat{\varrho}_t = \frac{\hat{\sigma}_t^2}{\mathrm{IV}_t^2}, \quad \varrho_t = \frac{\mathrm{RV}_t}{\mathrm{IV}_t^2},$$

  with mean $\overline{\mathrm{ES}}_\theta$ over the out-of-sample days.
- **The score** $\mathrm{ES}_{1,t}$ **is positive** only when $\hat{\sigma}_t^2$ and RV_t lie on opposite sides of the implied variance IV_t^2.
- **Compounding:** the premium bought or sold each day is a fixed fraction $f = 3\%$ of current wealth, so terminal wealth and the annualized log growth rate are

$$W_{\mathrm{oos}} = \prod_{t \in \mathcal{T}_{\mathrm{oos}}} (1 + f\, r_t), \qquad G = \frac{252}{T_{\mathrm{oos}}} \sum_{t \in \mathcal{T}_{\mathrm{oos}}} \log(1 + f\, r_t).$$

- At quoted bid/ask prices, r_t^{ba} replaces $r_t = q_t R_t$. Maximum drawdown $\max_t(1 - W_t / \max_{t' \le t} W_{t'})$, with W_t the wealth after day t.

7.3 Results

- **The unconditional short-straddle benchmark** ($q_t \equiv -1$; "Short straddle" in the tables): SR $= 0.20$ at mid-quotes and -0.27 at quoted bid/ask prices (Table 10). Its mean return is not significantly different from zero (HAC $t = 0.43$), and it loses money in 3 of the 5 calendar years: selling the closing straddle earns no statistically detectable variance risk premium in the out-of-sample period.
- **Sign strategies versus the short-straddle benchmark** (point estimates): $\Delta\text{SR} > 0$ for 112 of the 112 forecasts of the model set.
  - 20 of the 112 gains are significant at the 5% level (HAC t-statistic on the daily return differential);
  - all 20 use the full predictor set.
- **Highest Sharpe ratios** (point estimates):
  - closing-interval elastic net on the full predictor set: SR $= 1.92$ at mid-quotes and 1.43 at quoted bid/ask prices; its gain over the short-straddle benchmark is significant at the 5% level (HAC $t = 2.56$);
  - closing-interval lasso: 1.88 at mid-quotes, 1.40 at quoted bid/ask prices;
  - closing-interval LightGBM, the best of its specifications by out-of-sample SR (an ex post selection): 1.81;
  - MinGRU network on the full predictor set: 1.53 at mid-quotes, 1.05 at quoted bid/ask prices.
- **Full predictor set versus HAR-benchmark predictors:** the median ΔSR across the 51 estimator pairs is $+0.54$ (elastic net $+0.55$, lasso $+0.50$; Table 11). The gain is economically large but statistically significant in only 4 of the 51 pairs.
- **Position sizing,** median SR across the model set (Table 13): sign strategy 0.92, robust clip 1.34, rank 1.34.
  - $\text{SR}^{\text{clip}} > \text{SR}^{\text{sign}}$ for 98 of the 112 forecasts (point estimates; no test).
- **The value is concentrated in a few days** (Figure 2):
  - without its 10 best days, the mean return differential over the short-straddle benchmark, $\bar{\Delta}$, is negative for 102 of the 112 forecasts;
  - days on which the package expires worthless ($R_t = -1$, settlement pins) supply most of the return.
- **Statistical accuracy is a weak guide to economic value.** Spearman rank correlations across forecasts between average loss and the Sharpe ratio (negative: the more accurate forecasts earn the higher SR):

$$\rho_S(\overline{\text{QLIKE}}, \text{SR}) = -0.20, \qquad \rho_S(\overline{\text{ES}}_1, \text{SR}) = -0.55.$$

- **Robustness** (appendix):
  - shift test, $q_t = \text{sign}\big(\hat{y}^2_{t,c+j}\, s^2_{t,c} - \text{IV}^2_t\big)$, with the forecast of the interval j half hours away from the closing interval ($j < 0$: earlier on day t; $j > 0$: on day $t+1$): SR swings by about ± 1 across j, so differences of this size are sampling noise;
  - subsample splits around May 2022;
  - compounding: median terminal wealth W_{oos} of 4.4 at mid-quotes and 1.8 at quoted bid/ask prices; maximum drawdowns of up to 64% and 78%, respectively.

Table 10: Returns of the sign strategy

	Mean	SR		Hit	Long	
	return (%)	Mid-quotes	Bid–ask	rate (%)	days (%)	ΔSR
Short straddle	1.4	0.20	−0.27	61.8	0.0	–
	(0.43)					
Panel A: Closing-interval models						
OLS, HAR benchmark	9.7***	1.37	0.89	56.7	27.9	1.17*
	(2.83)					(1.85)
Elastic net	13.5***	1.92	1.43	57.7	28.3	1.72**
	(3.66)					(2.56)
Lasso	13.2***	1.88	1.40	57.6	29.6	1.67**
	(3.81)					(2.54)
Ridge	10.0***	1.42	0.93	57.5	27.1	1.21**
	(2.86)					(1.98)
LightGBM[a]	12.8***	1.81	1.33	56.9	30.5	1.61**
	(3.73)					(2.33)
XGBoost[a]	12.5***	1.77	1.27	58.7	27.8	1.56**
	(3.82)					(2.56)
Random forest[a]	11.2***	1.58	1.11	56.8	30.4	1.38**
	(3.38)					(2.07)
MinGRU[b]	10.8***	1.53	1.05	56.2	32.8	1.33**
	(3.09)					(2.02)
Panel B: Pooled intraday models						
Elastic net	10.6***	1.50	1.01	57.3	27.4	1.29*
	(2.81)					(1.90)
Lasso	8.5**	1.20	0.71	58.0	23.7	1.00*
	(2.45)					(1.66)
Ridge	10.2***	1.44	0.96	57.5	27.1	1.24*
	(2.78)					(1.93)

Notes: Daily returns of the straddle position over the out-of-sample period, 3 January 2020 to 30 April 2024 ($T_{\text{oos}} = 866$ trading days). The sign strategy holds $q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2)$ straddles; the short-straddle benchmark (Short straddle) holds $q_t \equiv -1$; every model except the OLS uses the full predictor set. Mean return is at mid-quotes, in percent of the premium; SR is the annualized Sharpe ratio at mid-quotes and at quoted bid/ask prices; hit rate and long days are the shares of days with $r_t > 0$ at mid-quotes and with $q_t = +1$. ΔSR is SR at mid-quotes minus that of the short-straddle benchmark (0.20); in parentheses, the HAC t-statistic of the mean return and, under ΔSR, the HAC t-statistic on the daily return differential (Newey–West, $q_{\text{NW}} = 14$; Section 5.5). *, ** and *** denote significance at the 10%, 5% and 1% levels.

[a] The specification with the highest SR at mid-quotes among the estimator's 15 on the full predictor set, so selected ex post on the out-of-sample period. LightGBM: random search, re-estimated every 10 trading days, selected on squared-error loss; XGBoost: random search, re-estimated daily, selected on squared-error loss; random forest: Optuna search (10 trials), re-tuned every 25 trading days, selected on squared-error loss.

[b] MinGRU network (ridge-initialized, trained and selected on QLIKE; Section 5.2).

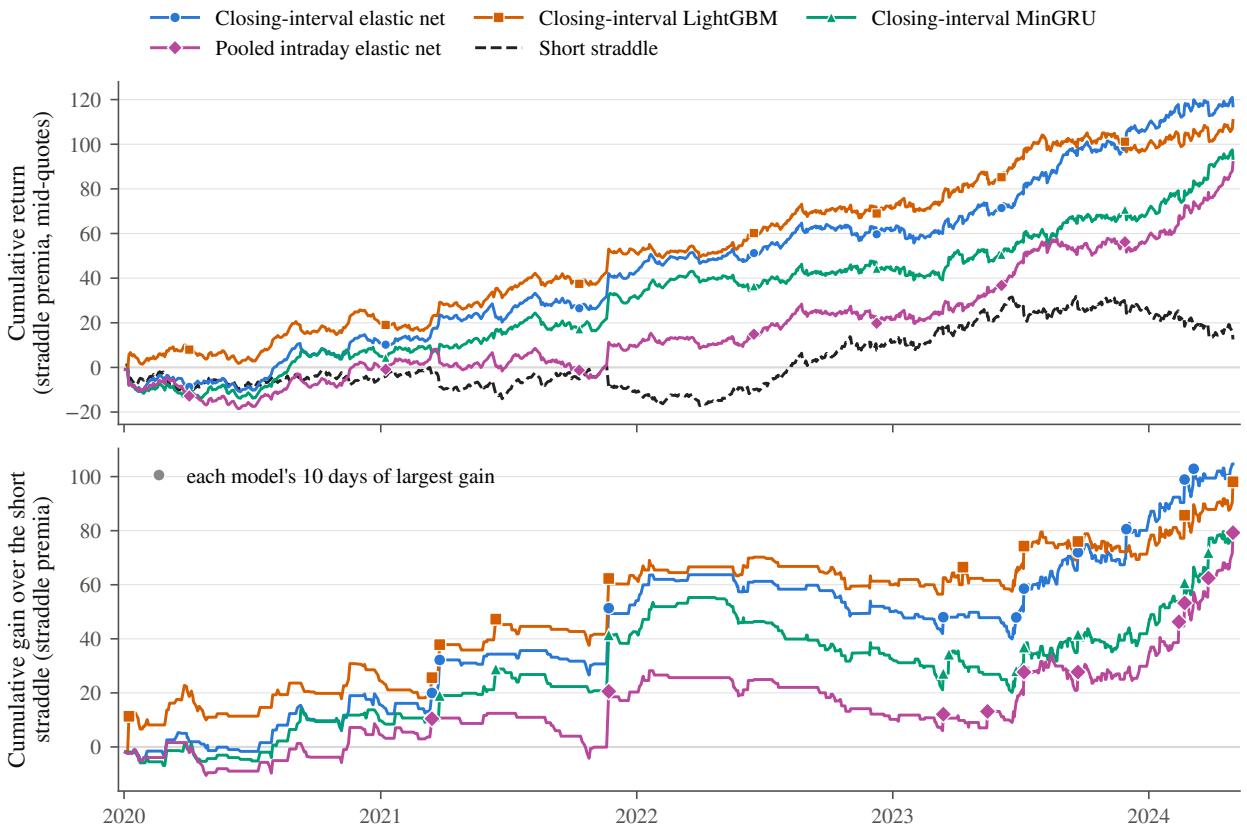


Figure 2: Cumulative returns of the sign strategy. The top panel plots the cumulative return at mid-quotes, $\sum_{t' \le t} r_{t'}$, in units of the straddle premium, for the short-straddle benchmark ($q_t \equiv -1$) and for one model of each class on the full predictor set: the closing-interval elastic net, LightGBM (the specification of Table 10) and MinGRU, and the pooled intraday elastic net. The bottom panel plots the cumulative gain over the short-straddle benchmark, $\sum_{t' \le t} \big(r_{t'} + R_{t'}\big)$; the markers are each model's 10 days of largest gain, all of them days with $q_t = +1$ and $R_t \ge 2.99$. Over the 866 trading days of the out-of-sample period the gain is +104.7, +98.1, +81.1 and +79.3 premia, in the order above, and without each model's 10 best days it is +15.6, +2.4, −6.2 and −6.3. Without their 10 best days, 102 of the 112 forecasts of the model set have a negative gain.

Table 11: Sharpe ratios: HAR benchmark versus full predictor set

	SR at mid-quotes		ΔSR	
	HAR benchmark	Full set	Mid-quotes	Bid–ask
Panel A: Closing-interval models				
Elastic net	1.37	1.92	0.55	0.54
			(1.34)	(1.35)
Lasso	1.37	1.88	0.50	0.51
			(1.47)	(1.48)
Ridge	1.37	1.42	0.05	0.04
			(0.09)	(0.10)
MinGRU[a]	1.04	1.53	0.49	0.49
			(0.95)	(0.95)
Panel B: Pooled intraday models				
Elastic net	0.89	1.50	0.61	0.60
			(1.21)	(1.20)
Lasso	0.69	1.20	0.51	0.50
			(1.19)	(1.19)
Ridge	0.91	1.44	0.53	0.52
			(1.12)	(1.12)
Panel C: Medians across pairs				
LightGBM (15)	0.73	1.33	0.60	0.60
			[0/15]	[0/15]
XGBoost (15)	0.80	1.48	0.55	0.54
			[3/15]	[3/15]
Random forest (15)	0.64	1.07	0.47	0.47
			[1/15]	[1/15]
All trees (45)	0.70	1.24	0.55	0.54
			[4/45]	[4/45]
Model set (51)	0.73	1.27	0.54	0.53
			[4/51]	[4/51]

Notes: Each pair is one model estimated on the HAR benchmark and on the full predictor set, with everything else unchanged, over the out-of-sample period ($T_{\text{oos}} = 866$ trading days). SR is the annualized Sharpe ratio of the sign strategy; ΔSR is the full-set minus the HAR-benchmark SR, at mid-quotes and at quoted bid/ask prices, with the HAC t-statistic on the daily return differential (Newey–West, $q_{\text{NW}} = 14$; Section 5.5) in parentheses. On the HAR benchmark the three closing-interval linear models hold the same position as the closing-interval OLS on every day, so they share its SR. Panel C gives medians across the closing-interval tree pairs (every hyperparameter specification in the model set) and across all pairs of the 112 forecasts of the model set, with the number of pairs whose differential is significantly positive at the 5% level in brackets (none is significantly negative at that level). *, ** and *** denote significance at the 10%, 5% and 1% levels.
[a] MinGRU network (ridge-initialized, trained and selected on QLIKE); not in the model set.

Table 12: Closing-interval versus pooled intraday models

	SR at mid-quotes		Pooled intraday minus closing-interval		
	Pooled intraday	Closing interval	ΔSR, mid-quotes	ΔSR, bid–ask	Δ QLIKE (%)
Panel A: Full predictor set					
Elastic net	1.50	1.92	−0.42	−0.42	8.5**
			(−0.88)	(−0.88)	(2.03)
Lasso	1.20	1.88	−0.68*	−0.68*	14.9***
			(−1.66)	(−1.67)	(3.09)
Ridge	1.44	1.42	0.03	0.02	7.2
			(0.07)	(0.06)	(1.61)
Panel B: HAR benchmark					
Elastic net	0.89	1.37	−0.49	−0.48	13.4***
			(−1.12)	(−1.12)	(3.45)
Lasso	0.69	1.37	−0.68*	−0.68*	13.0***
			(−1.69)	(−1.69)	(3.39)
Ridge	0.91	1.37	−0.46	−0.46	13.6***
			(−1.05)	(−1.04)	(3.37)

Notes: Each row compares the closing-interval model with the pooled intraday model of the same estimator and predictor set, both with hyperparameters selected on a pre-sample, over the out-of-sample period ($T_{\text{oos}} = 866$ trading days); the pooled intraday model is evaluated on the closing interval. SR is the annualized Sharpe ratio of the sign strategy; ΔSR is the pooled intraday minus the closing-interval SR, at mid-quotes and at quoted bid/ask prices, with the HAC t-statistic on the daily return differential (Newey–West, $q_{\text{NW}} = 14$; Section 5.5) in parentheses. Δ QLIKE is the pooled intraday model's mean QLIKE relative to the closing-interval model's, in percent, with the Diebold–Mariano statistic (Harvey–Leybourne–Newbold correction, Newey–West $q_{\text{NW}} = 6$) in parentheses; positive values favor the closing-interval model, as in Table 8. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Table 13: Sharpe ratios by position-sizing rule

	Sign	Robust clip	Full long, scaled short	Rank
Panel A: SR *at mid-quotes*				
Short straddle		0.34 under every rule		
Closing-interval models				
OLS, HAR benchmark	1.64	1.55	1.69	1.67
Elastic net	1.93	1.82	1.84	1.90
Lasso	1.76	1.75	1.74	1.85
Ridge	1.51	1.70	1.56	1.75
LightGBM	1.57	1.78	1.64	1.66
XGBoost	1.34	1.63	1.58	1.67
Random forest	1.20	1.28	1.28	1.26
MinGRU	1.59	1.74	1.57	1.79
Pooled intraday models				
Elastic net	1.91	1.75	1.72	1.69
Lasso	1.56	1.69	1.65	1.59
Ridge	1.73	1.79	1.68	1.73
Model set, median	0.92	1.34	1.12	1.34
Model set, number above sign	–	98	100	98
Panel B: SR *at quoted bid/ask prices*				
Short straddle		−0.03 under every rule		
Closing-interval models				
OLS, HAR benchmark	1.27	1.20	1.38	1.31
Elastic net	1.56	1.46	1.52	1.54
Lasso	1.40	1.39	1.42	1.49
Ridge	1.14	1.35	1.24	1.39
LightGBM	1.20	1.44	1.32	1.31
XGBoost	0.96	1.27	1.21	1.30
Random forest	0.84	0.93	0.95	0.90
MinGRU	1.22	1.39	1.24	1.44
Pooled intraday models				
Elastic net	1.54	1.38	1.40	1.32
Lasso	1.19	1.33	1.32	1.22
Ridge	1.36	1.43	1.35	1.35
Model set, median	0.56	0.99	0.79	0.98
Model set, number above sign	–	98	106	100

Notes: Annualized Sharpe ratios of the straddle position q_t under four sizing rules (Section 7.1), each a function of $\hat{\sigma}_t^2 - \mathrm{IV}_t^2$ and of its values on earlier days $t' < t$ only; the sign rule is the sign strategy, and the short-straddle benchmark (Short straddle) holds $q_t \equiv -1$ under every rule. The first 252 trading days of the out-of-sample period only estimate the scale of $|\hat{\sigma}_{t'}^2 - \mathrm{IV}_{t'}^2|$, so every rule is evaluated on the remaining 614 of the 866 trading days; at quoted bid/ask prices the bid–ask spread is paid on $|q_t|$ straddles. Every model except the OLS uses the full predictor set; the trees are the specifications of Table 10. The last two rows of each panel give the median across the 112 forecasts of the model set (the MinGRU is not in it) and the number of them whose SR is above that of their own sign strategy. Point estimates; no test is reported.

8 Variable importance

- **Models:** the closing-interval models of record on the full predictor set ($p = 628$).
  - Lasso and ridge with the penalty strength λ selected on a pre-sample: $\lambda = 10^{-2}$ and 10^{3} (scikit-learn parametrization). Penalty weight ω_j, the factor on predictor j's penalty term: $\underline{\omega} = 0.01$ on the HAR benchmark block, 1 on the exogenous predictors. Re-estimated every trading day; the scheduled re-selection of λ never replaced the pre-sample choice.
  - Not the earlier study's single-penalty specification (one λ for every predictor, re-selected every 250 trading days): the paper does not use it.
  - Tree ensembles at fixed hyperparameters, the best entries of the development-sample search (LightGBM and XGBoost: the top Optuna trial; random forest: trial 255); constant and duplicate predictors dropped in each estimation window.
- **Notation.**
  - Refit $k = 1, \ldots, K$, $K = 147$, one every 10 trading days. Refit k is estimated on $\mathcal{W}_{t_k}$, the 2,000 trading days before its first forecast day t_k, and forecasts its out-of-sample block $\mathcal{O}_k$, the trading days from t_k to the next refit (≤ 10). The blocks make up this section's out-of-sample period $\mathcal{O} = \bigcup_k \mathcal{O}_k$: $|\mathcal{O}| = 1{,}469$ trading days, 25 June 2018 to 30 April 2024.
  - Predictors $\boldsymbol{x}_t \in \mathbb{R}^p$; forecast $\hat{y}_t = \hat{f}_k(\boldsymbol{x}_t)$, $t \in \mathcal{O}_k$, on the scale of the target y_t, the periodicity-adjusted realized volatility of the closing half hour.
  - Predictor group $A \subseteq \{1, \ldots, p\}$: all predictors built from one series (its averages at every horizon, its missing-data and activity indicators). 49 groups: 47 exogenous series in 8 categories (Section 4), the HAR components, the calendar dummies.
  - Linear models: $\hat{\boldsymbol{\beta}}_k \in \mathbb{R}^p$, the coefficient estimates behind the forecast for t_k; $\hat{\boldsymbol{\beta}}_{k,A}$, its subvector on group A.
  - Trees: SHAP values $\phi_{t,j}$ (Lundberg and Lee, 2017) and SHAP interaction values $\phi_{t,jl}$ (Lundberg et al., 2020); they add up to the forecast,

$$\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}, \qquad \phi_{t,j} = \sum_{l=1}^{p} \phi_{t,jl}, \qquad \phi_{t,A} = \sum_{j \in A} \phi_{t,j}.$$

  - Path-dependent TreeSHAP (Lundberg et al., 2020): the training observations per node (node covers) stand in for the background sample.
  - Number triples below: LightGBM, XGBoost, random forest.
- **Lasso, aggregate:** the selection count of a group and of a predictor, the number of the K refits with a nonzero coefficient (Tables 15 and 16),

$$K_A = \sum_{k=1}^{K} \mathbf{1}\{\exists j \in A : \ \hat{\beta}_{k,j} \neq 0\}, \qquad K_j = \sum_{k=1}^{K} \mathbf{1}\{\hat{\beta}_{k,j} \neq 0\};$$

  K_A/K is the selection frequency, the share of estimation windows with a nonzero coefficient.
  - Per refit: 30 nonzero coefficients (median; range 27–35), of which 14 on exogenous predictors.
  - Predictors: 60 of 628 selected at least once, 29 in at least half of the refits.
  - Exogenous series: 27 of 47 selected at least once; 11 in at least half of the refits; 3 in every refit (r, $\mathrm{RAV}^{\mathrm{VW}}$, AC).

  - HAR benchmark block: selected in every refit ($\omega_j = 0.01$); 88 % of $\sum_j |\hat{\beta}_{k,j}|$ (median refit).
  - Random-probe benchmark: the continuous probe, an i.i.d. uniform noise predictor (defined under Known issues), is selected in 2.0 % of its fits; 54 predictors, in 23 exogenous series, are selected more often.

- **Lasso, time-varying:** the magnitude and the selection indicator of each group at each refit,

$$\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 = \sum_{j \in A} |\hat{\beta}_{k,j}|, \qquad \mathbf{1}\{\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 > 0\}$$

  (Table 16, lasso columns). Rank stability across refits: Kendall's coefficient of concordance W of the per-refit rankings by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ ($W = 1$: the same ranking at every refit; formula below), $W = 0.74$.
- **Ridge, aggregate:** the group's share of the absolute coefficient mass,

$$S_A = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}|}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}|}.$$

  - Scale check: only the averages of series values are robust-scaled.
    * Averages of series values (282 predictors): standard deviation over the estimation window 0.84 (median; 5th–95th percentiles: 0.03–1.76), interquartile range 0.94.
    * Averaged missing-data and activity indicators (324): standard deviation 0.022; calendar dummies 0.39; HAR components 1.07.
    * So the raw $|\hat{\beta}_{k,j}|$ are in different units. Check with the group's share of the absolute standardized coefficients,

$$S_A^{\mathrm{sd}} = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}},$$

      with $\mathrm{sd}_{k,j}$ the standard deviation of predictor j over $\mathcal{W}_{t_k}$ (Tables 15 and 16); Spearman rank correlation of S_A^{sd} with S_A across groups 0.85.
    * Leading groups on that scale: HAR components 30.6 %, calendar dummies 9.3 %, r 4.5 %.
  - Leading groups: calendar dummies 31.0 %, HAR components 22.8 %, r 3.6 %; HAR benchmark block 54 % (median refit).
  - Ridge sets no coefficient to zero: all 49 groups have $K_A = K$, so the selection frequency does not discriminate.
  - Random-probe benchmark: 107 predictors, in 38 exogenous series, have $|\hat{\beta}_{k,j}|$ above the continuous probe's (robust-scaled) in most probe fits.
- **Ridge, time-varying:** the per-refit share $S_{k,A} = \|\hat{\boldsymbol{\beta}}_{k,A}\|_1 / \sum_{A'} \|\hat{\boldsymbol{\beta}}_{k,A'}\|_1$, A' over all groups (Table 16, ridge columns, mean rank by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$); Kendall's $W = 0.85$.
- **Trees, per-refit ranking:** the group's mean absolute SHAP value on the refit's out-of-sample block, then its rank (Table 17, Figure 3),

$$I_{k,A} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,A}|, \qquad r_{k,A} = 1 + \#\{A' : I_{k,A'} > I_{k,A}\} \quad \text{(ties averaged)}.$$

- **Trees, global ranking:** aggregate the K rankings over time,

$$\bar{r}_A = \frac{1}{K} \sum_{k=1}^{K} r_{k,A}, \qquad \mathrm{BC}_A = \sum_{k=1}^{K} \max(0,\ 11 - r_{k,A}).$$

– BC_A: truncated Borda count (a full Borda count is affine in $\bar{r}_A$).
– Kendall's coefficient of concordance (Kendall and Babington Smith, 1939) over the K rankings of the n groups, with rank sums $R_A = \sum_k r_{k,A}$ and their mean $\bar{R}$:

$$W = \frac{12 \sum_A (R_A - \bar{R})^2}{K^2(n^3 - n) - K \sum_k \sum_{g \in \mathcal{E}_k} (|g|^3 - |g|)},$$

$\mathcal{E}_k$ the sets of tied (equally ranked) groups in refit k; $W = 1$: the same ranking at every refit.
– LightGBM: HAR components ($\bar{r}_A = 1.0$), r (2.9), RAV (4.8); Kendall's $W = 0.57$.
– XGBoost: HAR components (1.0), r (3.0), RAV (3.9); Kendall's $W = 0.60$.
– Random forest: HAR components (1.0), RAV (2.6), r (4.0); Kendall's $W = 0.62$.
– Mean rank vs truncated Borda count, Spearman rank correlation over groups: 0.90, 0.87, 0.92.
– Aggregate of the ranks vs rank of the aggregate, the mean absolute SHAP value over the out-of-sample period $\bar{I}_A = |\mathcal{O}|^{-1} \sum_{t \in \mathcal{O}} |\phi_{t,A}|$: 0.96, 0.97, 0.98.
– Predictor level ($n = 628$): Kendall's $W = 0.91$, 0.90, 0.93.
– Agreement of $\bar{r}_A$ across models (Spearman rank correlation): trees 0.83–0.96; tree vs linear 0.17–0.55; lasso vs ridge 0.47.

- **Interaction effects, global:** the mean absolute SHAP interaction value of two groups over the out-of-sample period (Table 18),

$$\phi_{t,AB} = \sum_{j \in A} \sum_{l \in B} (\phi_{t,jl} + \phi_{t,lj}), \qquad \bar{I}_{AB} = \frac{1}{|\mathcal{O}|} \sum_{t \in \mathcal{O}} |\phi_{t,AB}|, \quad A \neq B;$$

for two predictors, $|\phi_{t,jl} + \phi_{t,lj}|$, $j \neq l$.
– $\phi_{t,jl} = \frac{1}{2}(\phi_{t,l} \mid j \text{ present} - \phi_{t,l} \mid j \text{ absent})$, $j \neq l$; $\phi_{t,jj} = \phi_{t,j} - \sum_{l \neq j} \phi_{t,jl}$.
– Top pair: LightGBM HAR components × RAV (12.6); XGBoost HAR components × RAV (15.2); random forest HAR components × RAV (23.8); $\times 10^{-3}$, in units of y_t.
– Scale: largest $\bar{I}_A$ 195, 192, 224 ($\times 10^{-3}$); all series-pair $\bar{I}_{AB}$ summed, as % of $\sum_A \bar{I}_A$: 90, 74, 72.
– Within a group ($\sum_{j \neq l \in A} \phi_{t,jl}$): largest HAR components (18.7), HAR components (19.1), HAR components (34.3); $\times 10^{-3}$.
– Noise floor: the continuous probe's largest $\bar{I}$ with any group, median over probe fits: 0.49, 0.41, 0.50 ($\times 10^{-3}$); series pairs above it: 116, 115, 52 of 1,176.

- **Interaction effects, cross-check:** the H-statistic of Friedman and Popescu (2008) on each model's top 10 series pairs, from the partial dependence functions PD_S, $S \in \{A, B, AB\}$, $AB \equiv A \cup B$:

$$H^2_{AB} = \frac{\sum_t [\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)]^2}{\sum_t \mathrm{PD}_{AB}(\boldsymbol{x}_t)^2}, \qquad \mathrm{PD}_S(\boldsymbol{x}) = \frac{1}{|\mathcal{H}_k|} \sum_{t' \in \mathcal{H}_k} \hat{f}_k(\boldsymbol{x}_S, \boldsymbol{x}_{t',-S}),$$

$t \in \mathcal{H}_k$; each PD_S centered over $\mathcal{H}_k$.
– Points: $\mathcal{H}_k$, the 100 trading days from t_k on, both evaluation points and background sample, at 15 refits (every tenth); sums pooled over refits.

  - The denominator contains the HAR components' own partial dependence whenever they are in the pair. A second normalization, by the smaller single-group partial dependence:

$$H^2_{\min,AB} = \frac{\sum_t \left[\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)\right]^2}{\min\{\sum_t \mathrm{PD}_A(\boldsymbol{x}_t)^2,\ \sum_t \mathrm{PD}_B(\boldsymbol{x}_t)^2\}}.$$

  - Top pair, H^2 (%): 0.33, 0.37, 0.85; over the top 10: 0.02–0.33, 0.02–0.68, 0.02–0.85.
  - Top pair, $H^2_{\min}$: 0.19, 0.39, 1.50; over the top 10: 0.05–0.54, 0.08–0.80, 0.17–5.46.
  - Top predictor pair, H^2 (%) and $H^2_{\min}$: LightGBM 0.36, 0.02; XGBoost 0.39, 0.02; random forest 1.16, 0.06.
- **Interaction effects, time-varying:** $\phi_{t,AB}$ on each refit's out-of-sample block, ranked per refit, aggregated as the SHAP values are,

$$I_{k,AB} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,AB}| \ \rightarrow\ r_{k,AB} \ \rightarrow\ \bar{r}_{AB},\ W.$$

  - Kendall's W over each model's top 50 pairs: 0.45, 0.47, 0.57.
  - Mean rank vs $\bar{I}_{AB}$ (Spearman rank correlation, all pairs): 0.99, 0.99, 1.00.
- **Known issues, reconciled:**
  - Mean decrease in impurity (MDI) favors predictors with many distinct values (Strobl et al., 2007): MDI is not used; SHAP values and coefficients only.
  - Random probes, at every third refit (49 fits): three i.i.d. noise predictors, continuous (U(0, 1)), discrete uniform on 20 levels, and binary (Bernoulli, probability one half).
    - Lasso: the probes are selected in 1, 0, 0 of 49 fits (robust-scaled, $\omega_j = 1$).
    - Ridge: 107 predictors have $|\hat{\beta}_{k,j}|$ above the continuous probe's in most probe fits.
    - Trees: the continuous probe's median rank among 631 predictors by mean $|\phi_{t,j}|$: 83, 110, 97; exogenous series with a predictor above it in most probe fits: 27, 35, 29.
  - Dilution across correlated predictors: $\phi_{t,A}$ and $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ sum over the group's predictors, so credit split among a series' averages at different horizons stays inside the group.
  - The earlier study's single-penalty lasso (λ re-selected): median 23 nonzero coefficients per refit; 18 exogenous series selected in at least half of the refits; Spearman rank correlation of its K_A with that of the lasso of record 0.70.
  - Re-estimation: LightGBM reproduces the archived refits exactly; XGBoost and the random forest differ across library versions (maximum absolute forecast difference 0.17 and 0.11, in units of y_t). SHAP values: the archived refits'; interaction values: the re-estimated ones' (Spearman rank correlation of $\bar{r}_A$, re-estimated vs archived: 1.000, 0.993, 0.996).
- **Compute** (one laptop, 11 processes):
  - Tree re-estimation and SHAP interaction values: 133 minutes. Per refit: random forest estimation 212 s, its SHAP and interaction values 182 s; LightGBM 0.9 s, XGBoost 0.4 s.
  - Linear coefficient paths and probes: 7 minutes; H^2: 7 minutes.

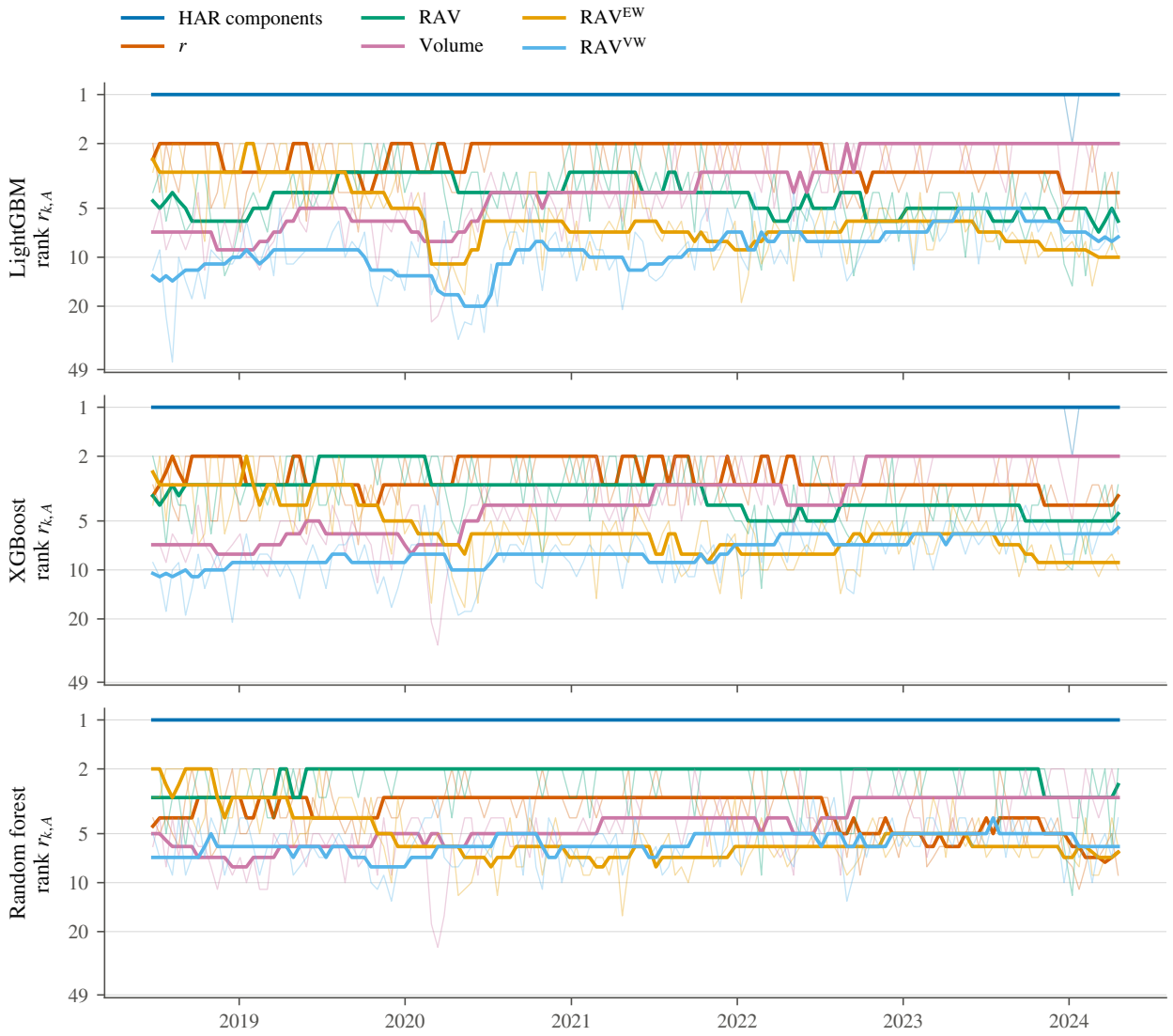


Figure 3: Mean absolute SHAP rank of the leading predictor groups over time. At refit k (one every 10 trading days), $r_{k,A}$ is the rank of group A among the 49 groups by its mean absolute SHAP value on the out-of-sample block $\mathcal{O}_k$ (1 = largest; log scale), plotted at the block's first day t_k. Thin lines: one point per refit; thick lines: centered rolling median over 11 refits (about 110 trading days). Groups: the six with the lowest mean rank $\bar{r}_A$ averaged over the three tree ensembles; Kendall's W is in Table 15.

Table 14: Importance of the HAR components in the tree models

Model	MDI share (%)	SHAP share (%)	Permutation ΔQLIKE (%)	Close-interaction share (%)	MDI rank
LightGBM	57 → 52	47 → 44	237 → 200	22	1 → 1
XGBoost	69 → 54	55 → 47	224 → 181	22	1 → 1
Random forest	62 → 59	58 → 61	290 → 271	54	1 → 1

Notes: Draft, measures under review. Closing-interval tree models on the full predictor set, first pass (old) → current design (new); the first pass includes the 12 interactions of the HAR components with the opening- and closing-interval dummies and no column screening, the current design drops those interactions and screens out constant and duplicate columns in every estimation window. MDI and SHAP: the HAR components' share of the model's total; permutation ΔQLIKE: the increase in QLIKE when their values are replaced by rows drawn from the estimation window, in percent of the model's out-of-sample loss. Close-interaction share: the part of the old MDI carried by the 6 closing-interval interactions, which at the closing interval are copies of the HAR components; MDI rank: rank among the series. XGBoost and the random forest also changed hyperparameters between the passes, so their changes mix design, screening and hyperparameters.

Table 15: Variable importance by model

Panel A: Linear models

	Lasso		Ridge		Ridge, standardized	
Rank	Group	K_A	Group	S_A (%)	Group	S_A^{sd} (%)
1	Calendar dummies	147	Calendar dummies	31.0	HAR components	30.6
2	HAR components	147	HAR components	22.8	Calendar dummies	9.3
3	r	147	r	3.6	r	4.5
4	RAV^{VW}	147	RAV^{VW}	2.2	RAV^{VW}	3.7
5	AC	147	AC	2.1	$\text{EffSpread}^{\text{VW}}$	2.5
Kendall's W		0.74		0.85		–
Series above probe		23		38		–

Panel B: Tree ensembles

	LightGBM		XGBoost		Random forest	
Rank	Group	$\bar{r}_A$	Group	$\bar{r}_A$	Group	$\bar{r}_A$
1	HAR components	1.0	HAR components	1.0	HAR components	1.0
2	r	2.9	r	3.0	RAV	2.6
3	RAV	4.8	RAV	3.9	r	4.0
4	Volume	4.9	Volume	4.6	Volume	5.1
5	RAV^{EW}	6.7	RAV^{EW}	6.7	RAV^{EW}	5.7
Kendall's W		0.57		0.60		0.62
Series above probe		27		35		29

Notes: Closing-interval models on the full predictor set ($p = 628$ predictors in 49 predictor groups), $K = 147$ refits, one every 10 trading days, over the out-of-sample period 25 June 2018 to 30 April 2024 ($|\mathcal{O}| = 1{,}469$ trading days). Each panel lists the five leading groups by the measure in the column header (Section 8): the lasso's selection count K_A (refits with a nonzero coefficient on group A, of K; ties broken by the lasso's coefficient share), the ridge's share S_A of the absolute coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}|$ and its share S_A^{sd} of the absolute standardized coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}| \, \text{sd}_{k,j}$, and the tree ensembles' mean rank $\bar{r}_A$ by mean absolute SHAP value (1 = most important). Kendall's W is the concordance of the K per-refit rankings ($W = 1$: the same ranking at every refit). Series above probe: exogenous series with a predictor that beats the continuous noise probe ε^{U}, added in 49 probe fits (lasso: selected in a larger share of refits than the probe in its fits; ridge and trees: a larger $|\hat{\beta}_{k,j}|$ or mean $|\phi_{t,j}|$ than the probe's in at least half of the probe fits). Group symbols as in Table 30.

Table 16: Lasso selection frequency and ridge coefficient shares

	Lasso			Ridge		
Predictor group	K_A	Nonzero	S_A (%)	S_A (%)	S_A^{sd} (%)	$\bar{r}_A$
Panel A: HAR benchmark block						
Calendar dummies	147	10.0	53.2	31.0	9.3	1.0
HAR components	147	5.9	34.6	22.8	30.6	2.0
Panel B: Exogenous series						
r	147	3.0	5.4	3.6	4.5	3.0
$\mathrm{RAV}^{\mathrm{VW}}$	147	1.0	2.6	2.2	3.7	4.6
AC	147	1.3	1.2	2.1	2.2	5.0
$\mathrm{RS}^{+,\mathrm{EW}}$	126	1.2	0.6	1.1	1.9	17.8
$\mathrm{EffSpread}^{\mathrm{VW}}$	119	0.8	0.2	1.3	2.5	13.1
$\mathrm{TO}_{\mathrm{sell}}^{\mathrm{VW}}$	109	0.7	1.0	1.2	1.4	16.5
RQ	107	0.8	0.2	1.2	2.2	15.8
$\mathrm{RQ}^{\mathrm{VW}}$	106	0.7	0.1	0.7	1.2	34.2
$\mathrm{RS}^{+,\mathrm{VW}}$	105	0.8	0.2	1.0	1.7	21.2
$\mathrm{OFI}^{\mathrm{EW}}$	83	0.8	0.1	1.5	1.5	10.4
VIX	83	0.6	0.1	1.7	2.2	8.1
$\mathrm{RQ}^{\mathrm{EW}}$	65	0.5	0.1	0.7	1.1	33.6
$\mathrm{EffSpread}^{\mathrm{EW}}$	56	0.5	0.0	1.0	1.6	23.6
N	45	0.3	0.0	0.4	0.4	44.7
$\mathrm{FOMC}^{\mathrm{day}}$	42	0.3	0.0	0.4	0.6	44.4
$\mathrm{Spread}^{\mathrm{VW}}$	35	0.2	0.1	0.8	0.7	29.7
$\mathrm{BV}^{\mathrm{EW}}$	26	0.2	0.1	0.6	1.1	37.2
BV	25	0.2	0.0	0.5	1.0	41.2

Notes: Lasso and ridge on the full predictor set, $K = 147$ refits, one every 10 trading days, over the out-of-sample period 25 June 2018 to 30 April 2024 ($|\mathcal{O}| = 1{,}469$ trading days); penalty weight $\omega_j = \underline{\omega}$ on the HAR benchmark block (Panel A). Panel B lists the 18 exogenous series with the largest lasso selection count K_A, the number of refits with a nonzero coefficient on group A (ties broken by the lasso's S_A; measures defined in Section 8); 20 of the 49 groups are never selected. Nonzero: mean number of the group's predictors with $\hat{\beta}_{k,j} \neq 0$, of 12 per exogenous series (18 with activity averages), 6 HAR components and 16 calendar dummies (five of them constant or duplicated at the closing interval, so never in a fit). S_A: the group's share of the absolute coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}|$; S_A^{sd}: its share of the absolute standardized coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}$, which put the robust-scaled series averages and the unscaled indicator averages and dummies on one scale. $\bar{r}_A$: mean over refits of the group's rank by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ (1 = largest).

Table 17: Tree ensembles: mean SHAP rank by predictor group

	LightGBM		XGBoost		Random forest	
Predictor group	$\bar{r}_A$	Top 5 (%)	$\bar{r}_A$	Top 5 (%)	$\bar{r}_A$	Top 5 (%)
Panel A: HAR benchmark block						
HAR components	1.0	100.0	1.0	100.0	1.0	100.0
Calendar dummies	7.9	36.7	8.1	31.3	15.1	7.5
Panel B: Exogenous series						
r	2.9	96.6	3.0	97.3	4.0	82.3
RAV	4.8	70.7	3.9	87.1	2.6	96.6
Volume	4.9	70.7	4.6	74.8	5.1	69.4
RAV^{EW}	6.7	38.8	6.7	38.1	5.7	46.3
RAV^{VW}	10.0	15.0	8.6	12.9	6.1	45.6
Sent	11.9	15.6	11.6	17.0	11.3	19.0
AC	11.9	3.4	11.6	4.8	16.1	2.0
OFI^{EW}	14.6	13.6	14.8	10.9	12.9	8.8
RSK	17.9	2.7	16.5	0.7	15.4	2.0
NSent	17.1	6.8	16.2	4.1	17.6	3.4
$\text{TO}_{\text{sell}}^{\text{VW}}$	20.0	6.8	16.4	8.2	18.3	6.8
OFI^{VW}	25.8	0.7	26.0	2.0	18.4	1.4
Attn	22.7	1.4	24.4	0.7	23.7	0.0
Kendall's W	0.57		0.60		0.62	
Series above probe	27		35		29	

Notes: Path-dependent TreeSHAP values of Lundberg et al. (2020) for the closing-interval tree ensembles on the full predictor set, $K = 147$ refits, one every 10 trading days, over the out-of-sample period 25 June 2018 to 30 April 2024 ($|\mathcal{O}| = 1{,}469$ trading days). At refit k, group A is ranked among the 49 groups by its mean absolute SHAP value on the out-of-sample block, $I_{k,A} = |\mathcal{O}_k|^{-1} \sum_{t \in \mathcal{O}_k} |\phi_{t,A}|$ (rank $r_{k,A}$, 1 = largest). $\bar{r}_A$: mean of $r_{k,A}$ over the refits; Top 5: share of refits with $r_{k,A} \leq 5$. Rows: the 15 groups with the lowest $\bar{r}_A$ averaged over the three models. Kendall's W and series above probe as in Table 15; measures defined in Section 8.

Table 18: Tree ensembles: SHAP interaction effects

Pair	$\bar{I}_{AB}$ ($\times 10^3$)	$\bar{r}_{AB}$	Top 10 (%)	H^2 (%)	$H^2_{\min}$
Panel A: LightGBM					
Group pairs					
HAR components × RAV	12.64	1.7	100.0	0.33	0.19
HAR components × r	10.95	2.4	99.3	0.29	0.09
HAR components × Volume	6.62	6.3	87.8	0.05	0.05
HAR components × RAV^{EW}	5.46	6.3	88.4	0.05	0.12
HAR components × Sent	4.70	14.4	67.3	0.06	0.17
Predictor pairs					
$\bar{y}_t^{(1)} \times \bar{y}_t^{(5)}$	10.25	–	–	0.36	0.02
$\bar{y}_t^{(1)} \times \bar{y}_t^{(25)}$	4.63	–	–	0.10	0.02
$\bar{y}_t^{(5)} \times \text{RAV}_t^{(1)}$	4.53	–	–	0.67	0.42
Probe floor	0.49				
Panel B: XGBoost					
Group pairs					
HAR components × RAV	15.23	1.6	100.0	0.37	0.39
HAR components × r	11.03	2.5	100.0	0.68	0.21
HAR components × Volume	7.40	6.0	89.8	0.08	0.08
HAR components × RAV^{EW}	5.92	5.3	93.2	0.06	0.17
HAR components × Sent	4.87	11.8	69.4	0.09	0.25
Predictor pairs					
$\bar{y}_t^{(1)} \times \bar{y}_t^{(5)}$	9.95	–	–	0.39	0.02
$\bar{y}_t^{(1)} \times \bar{y}_t^{(25)}$	4.99	–	–	0.14	0.02
$\bar{y}_t^{(1)} \times \text{RAV}_t^{(5)}$	4.94	–	–	0.28	0.43
Probe floor	0.41				
Panel C: Random forest					
Group pairs					
HAR components × RAV	23.85	1.1	100.0	0.85	1.50
HAR components × r	9.13	3.5	99.3	0.20	0.48
HAR components × Volume	8.52	3.6	98.6	0.05	0.17
HAR components × RAV^{EW}	6.87	4.6	98.6	0.06	0.33
HAR components × RAV^{VW}	6.59	4.7	98.0	0.05	0.26
Predictor pairs					
$\bar{y}_t^{(1)} \times \bar{y}_t^{(5)}$	25.31	–	–	1.16	0.06
$\bar{y}_t^{(1)} \times \text{RAV}_t^{(1)}$	8.67	–	–	0.89	1.48
$\bar{y}_t^{(1)} \times \text{RAV}_t^{(5)}$	6.74	–	–	0.32	1.04
Probe floor	0.50				

Notes: SHAP interaction values $\phi_{t,jl}$ of Lundberg et al. (2020) for the re-estimated closing-interval tree ensembles on the out-of-sample blocks of the $K = 147$ refits ($|\mathcal{O}| =$ 1,469 trading days). Group pairs: the five with the largest $\bar{I}_{AB}$, the mean of $|\phi_{t,AB}|$ over $\mathcal{O}$ (Section 8) ($\times 10^3$, on the scale of y_t); predictor pairs: the three with the largest mean $|\phi_{t,jl} + \phi_{t,lj}|$. $\bar{r}_{AB}$ and Top 10: the pair's mean rank among the 1,176 group pairs over the refits and the share of refits in which it ranks in the top ten. H^2: the statistic of Friedman and Popescu (2008), in percent; $H^2_{\min}$: the same numerator over the smaller single-group partial-dependence variance; both pooled over 15 refits and computed for each model's top 10 group pairs and top five predictor pairs (–: not computed). Probe floor: median over the 49 probe fits of the largest $\bar{I}$ between the continuous noise probe ε^{U} and a group. Kendall's W of the per-refit rankings of each model's top 50 pairs: 0.45 (LightGBM), 0.47 (XGBoost) and 0.57 (random forest).

9 Conclusion

- **Exogenous predictors add to the HAR benchmark, consistently but modestly.**
  - $\overline{\text{QLIKE}}$ falls in most pairs and SR rises in nearly all.
  - Few gains are individually statistically significant over $T_{\text{oos}} = 866$ trading days.
- **The closing-interval model is more accurate than the pooled intraday model,** significantly so on the HAR benchmark.
  - The difference in the straddle strategy's SR is not statistically significant.
- **A lower QLIKE does not imply a higher Sharpe ratio.** The strategy uses $\hat{\sigma}_t^2$ only through the side of implied variance on which it falls:

$$q_t = \text{sign}(\hat{\sigma}_t^2 - \text{IV}_t^2).$$

- **The straddle strategy measures economic value; it is not a claim of implementable profits.**
  - Every forecast earns a higher SR than the unconditional short-straddle benchmark ($q_t \equiv -1$).
  - Its profits are concentrated in a small number of large-move days.
  - At quoted bid/ask prices, rather than mid-quotes, the SR is materially lower.
- **A recurrent network (MinGRU) matches the linear models** once regularized like them:
  - initialized at the ridge solution;
  - exogenous predictors shrunk more heavily than the HAR benchmark block, through weight decay in proportion to the linear penalty weights ω_j;
  - trained and selected on QLIKE, the evaluation loss, $\min_{\boldsymbol{\theta}} \sum_{t' \in \mathcal{W}_t} \text{QLIKE}(\tilde{y}_{t'}^2, \hat{y}_{t'}^2(\boldsymbol{\theta}))$ (up to the floor and clamp of Section 5.2), with $\tilde{y}_{t'}$ the winsorized target and t' the trading days of the estimation window; QLIKE is scale-free, so this is the same loss on the variance scale.
- **Limitations:**
  - $T_{\text{oos}} = 866$ trading days;
  - one index and one intraday interval;
  - no delta hedging and no price-impact model;
  - tree-ensemble forecasts depend on the CPU instruction set and are not bit-for-bit reproducible across machines.
- **Future work:**
  - evaluate the expirations after April 2024 out of sample;
  - density forecasts of closing-interval variance;
  - a second market;
  - forecast origins, and so straddle entries, earlier in the trading day;
  - transaction costs calibrated to executed prices.

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A Appendix structure

- **A. Data and predictors:**
  - sources and spans;
  - predictor dictionary: 47 exogenous series → 628 predictors in the full predictor set;
  - missing and stale data;
  - removal of duplicate predictors; the window filter (constant and duplicate predictors dropped in each estimation window $\mathcal{W}_t$);
  - SPXW option chain; implied-volatility inversion;
  - FOMC calendar.
- **B. Hyperparameter selection by model:**
  - linear models: penalty strength λ, elastic-net mixing α; block-penalized ridge, penalty weights ω_j;
  - tree ensembles: fixed hyperparameters, Optuna budgets, selected trials; boosting rounds $b = 1, \ldots, B$, learning rate η;

  - re-selection frequency, on the estimation sample $\mathcal{W}_t^{\text{est}}$, gap $\mathcal{W}_t^{\text{gap}}$ and validation sample $\mathcal{W}_t^{\text{val}}$; re-estimation frequency;
  - network: the hyperparameter grid Θ (look-back length ℓ, hidden width, learning rate, weight decay ν, ν_{enc}, ν_{lin}) and the selection on $\mathcal{W}_t^{\text{val}}$.
- **C. Robustness, statistical:**
  - full model table, tree ensembles included;
  - Diebold–Mariano and HAC conventions (DM with the Harvey–Leybourne–Newbold correction; Newey–West lag truncation q_{NW});
  - model confidence set, superior predictive ability, StepM;
  - regime and break tests;
  - closing-interval vs pooled intraday models;
  - remaining variance through the trading day.
- **D. Robustness, trading:**
  - date-shift test;
  - Murphy diagrams (elementary scores); subsample splits;
  - P&L decomposition;
  - compounding;
  - position sizing;
  - strategy variations; other forecast origins; month-ends.
- **E. Reproducibility:**
  - software lock files;
  - hashes of the design matrices;
  - computing environment;
  - CPU classes;
  - compute time;
  - the generator script behind every reported number.
- **F. Network implementation:**
  - native PufferLib 5.0 MinGRU trainer: logistic function $\Lambda(a) = 1/(1+e^{-a})$ in the update gate $\boldsymbol{z}_\tau$ and the highway gate $\boldsymbol{\psi}_\tau$; look-back ℓ; weight decay ν;
  - float64 reference checks; byte-identity replication checks;
  - scripts for the rolling-window out-of-sample exercise.

Table 19: Decomposition of the closing-interval results

	QLIKE		Mid-quotes		Bid–ask	Same
Change	Δ (%)	DM	ΔSR	t	ΔSR	sign (%)
Panel A: Earlier draft's model: block-penalized → single-penalty ridge, pooled intraday						
Full predictor set	−4.3*	(−1.73)	0.16	(0.34)	0.16	88
Panel B: Ridge: pooled intraday → closing-interval						
HAR benchmark	−12.0***	(−3.37)	0.46	(1.05)	0.46	88
Full predictor set	−6.7	(−1.61)	−0.03	(−0.07)	−0.02	86
Panel C: Closing-interval ridge: predictor set						
HAR benchmark → full predictor set	0.8	(0.18)	0.05	(0.09)	0.04	86
Panel D: Full predictor set: ridge → other model (trees re-estimated every 10 days)						
Lasso	−4.5	(−1.61)	0.46	(1.28)	0.47	92
Elastic net	−1.7	(−1.09)	0.50	(1.59)	0.50	93
LightGBM, fixed hyperparameters	0.8	(0.23)	−0.05	(−0.13)	−0.05	85
XGBoost, fixed hyperparameters	1.6	(0.45)	0.13	(0.35)	0.13	85
Random forest, fixed hyperparameters	3.2	(0.70)	−0.15	(−0.34)	−0.13	85
LightGBM, random search	3.1	(0.69)	0.39	(0.93)	0.40	85
XGBoost, random search	5.6	(1.26)	0.26	(0.46)	0.26	84
Random forest, random search	11.3**	(1.96)	−0.30	(−0.81)	−0.29	86
Panel E: Dropping the 12 duplicated HAR interactions						
Closing-interval ridge	0.0	–	0.00	–	0.00	100
Panel F: Within-window screening (LightGBM): screened minus unscreened						
Fixed hyperparameters, every 10 days	−1.0	(−1.04)	−0.21	(−0.69)	−0.21	96
Fixed hyperparameters, daily	1.4	(1.36)	−0.23	(−0.55)	−0.23	94
Random search, every 10 days	2.8	(1.42)	−0.08	(−0.32)	−0.08	94
Random search, daily	0.2	(0.12)	−0.05	(−0.17)	−0.05	94
Panel G: Re-estimation frequency (LightGBM): every 10 trading days → daily						
Fixed hyperparameters	−2.2	(−1.12)	−0.04	(−0.08)	−0.04	93
Panel H: Tuning of LightGBM (re-estimated daily); Optuna re-tuned daily or every 250 days						
Fixed → random search	3.0	(1.12)	0.28	(0.58)	0.28	91
Fixed → Optuna, daily	7.3***	(2.83)	−0.10	(−0.31)	−0.10	91
Optuna, every 250 days → daily	2.5	(0.75)	−0.32	(−0.65)	−0.33	90

Notes: Each row compares two forecasts that differ in one specification choice ("to" minus "from"), from the earlier draft's pooled intraday forecast to the closing-interval specification and through the tuning of the tree models, over the 866 trading days of the out-of-sample period. QLIKE Δ (%): $100\,(\overline{\text{QLIKE}}_{\text{to}}/\overline{\text{QLIKE}}_{\text{from}}-1)$, with the Diebold–Mariano statistic in parentheses (Harvey–Leybourne–Newbold correction, Newey–West lag $q_{\text{NW}}=6$; –: not stored). ΔSR: change in the annualized Sharpe ratio of the sign strategy, at mid-quotes with the HAC t-statistic on the daily return differential in parentheses ($q_{\text{NW}}=14$) and at quoted bid–ask prices; same sign: share of trading days on which the two forecasts imply the same position q_t. *, ** and *** denote significance at the 10%, 5% and 1% levels, from the DM statistic for Δ and from the HAC t for ΔSR. "Fixed": fixed hyperparameters; "every 10 days" and "daily" give the re-estimation frequency in Panels D and F and Optuna's re-tuning frequency in Panel H (Section 5.4).

Table 20: Re-estimation frequency and random search

	Re-estimated daily minus every 10 days				Random search	
	Fixed hyperparameters		Random search		minus fixed	
Model	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR
Panel A: HAR benchmark						
LightGBM	−0.7	0.06	0.2	−0.31	6.2	−0.15
XGBoost	0.1	−0.25	−0.1	−0.42	9.7	−0.38
Random forest	−1.4	−0.31	0.2	0.03	10.0	0.01
Panel B: Full predictor set						
LightGBM	−2.2	−0.04	−1.5	−0.20	3.0	0.28
XGBoost	−3.3	−0.43	−2.0	0.09	5.4	0.65
Random forest	−1.4	−0.43	−3.7	0.06	5.4	0.33

Notes: Closing-interval tree models. Columns 2–5: re-estimated daily minus re-estimated every 10 trading days, with fixed hyperparameters and with the random search; columns 6–7: the random search minus the fixed hyperparameters, both re-estimated daily. ΔQLIKE: the difference in mean QLIKE, in percent of the second forecast's; ΔSR: the difference in the annualized Sharpe ratio of the sign strategy at mid-quotes. Point estimates over the 866 trading days of the out-of-sample period; HAR benchmark 22 columns, full predictor set 628. Tuning procedures: Section 5.4.

Table 21: Within-window column screening

	Re-estimated every 10 trading days		Re-estimated daily	
Model, tuning	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR
Panel A: HAR benchmark				
LightGBM, fixed hyperparameters	0.4	−0.17	−0.1	−0.12
XGBoost, fixed hyperparameters	−0.6	−0.04	0.7	−0.08
Random forest, fixed hyperparameters	0.3	0.05	0.1	−0.10
LightGBM, random search	0.0	−0.13	−0.1	0.06
Panel B: Full predictor set				
LightGBM, fixed hyperparameters	−1.0	−0.21	1.4	−0.23
XGBoost, fixed hyperparameters	0.9	−0.22	0.8	−0.30
Random forest, fixed hyperparameters	0.0	−0.18	−0.4	−0.34
LightGBM, random search	2.8	−0.08	0.2	−0.05

Notes: Screened minus unscreened forecasts of the closing-interval tree models, by tuning and re-estimation frequency. At every estimation, the screen drops the columns that are constant on the estimation window or exact copies of an earlier column (Section 4). Cells as in Table 20. XGBoost and the random forest have no unscreened random-search run, so those rows are omitted; the unscreened runs were not restricted to a single CPU class.

Table 22: Hyperparameter re-tuning frequency

		Re-tuned more often minus every 250 trading days					
	QLIKE,	Daily		Every 5 days		Every 25 days	
Model	250 days	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR
Panel A: HAR benchmark							
LightGBM	0.1100	−4.1	−0.13	−5.0	−0.62	−4.3	−0.26
XGBoost	0.1092	−5.1	−0.42	−6.3	−0.42	−5.5	−0.55
Random forest	0.1060	−4.3	0.00	−4.3	−0.24	−4.1	−0.20
Panel B: Full predictor set							
LightGBM	0.1006	2.5	−0.32	0.8	−0.59	5.8	0.23
XGBoost	0.1013	0.5	−0.03	−0.7	−0.08	2.4	−0.01
Random forest	0.1027	−1.5	−0.35	−1.5	−0.08	1.0	0.08

Notes: Optuna re-tuned every 1, 5 or 25 trading days minus Optuna re-tuned every 250 trading days; every specification is re-estimated daily. At each tuning date, a new tree-structured Parzen estimator study runs 50 trials, starting from the fixed hyperparameters, and keeps the configuration with the lowest validation MSE (Section 5.4). QLIKE, 250 days: the mean QLIKE of the specification re-tuned every 250 trading days. Other cells as in Table 20.

Table 23: Adequacy of the Optuna trial budget

	Best of 50 trials			Validation MSE reduction (%)			Changed
Model	Median trial	Fixed (%)	Last ten (%)	From 10	From 25 mean / median	From 40	selection (%)
Panel A: HAR benchmark							
LightGBM	35	0.1	36.8	5.22 / 4.42	1.66 / 0.99	0.50 / 0.00	72
XGBoost	34	0.3	36.5	5.22 / 4.36	1.95 / 1.04	0.66 / 0.00	70
Random forest	32	1.0	26.8	2.69 / 1.95	0.62 / 0.28	0.16 / 0.00	63
Panel B: Full predictor set							
LightGBM	33	0.7	31.9	5.96 / 4.86	2.15 / 1.13	0.69 / 0.00	66
XGBoost	33	0.2	33.6	5.91 / 4.90	2.15 / 1.18	0.71 / 0.00	68
Random forest	33	1.0	31.2	3.22 / 2.32	0.81 / 0.44	0.23 / 0.00	68

Notes: Position of the best of 50 trials, and the gain from the last trials, over all 1,469 tuning dates of each specification (Section 5.4). Median trial: the median index of the trial that attains the best of 50; fixed: the share of tuning dates at which that trial is the fixed-hyperparameter configuration (trial 0); last ten: the share at which it falls in the last ten trials (20% if its position were exchangeable). Validation MSE reduction: the relative reduction in the best validation MSE from the first 10, 25 or 40 trials to all 50, mean / median across tuning dates. Changed selection: the share of tuning dates at which the best of the first 25 trials differs from the best of 50.

Table 24: Optuna trial budget, out of sample

	Best of 50 trials minus best of the first			
	10 trials		25 trials	
Model, re-tuning frequency	ΔQLIKE (%)	ΔSR	ΔQLIKE (%)	ΔSR
Panel A: HAR benchmark				
LightGBM, re-tuned daily	0.4	0.21	−0.3	0.49
XGBoost, re-tuned daily	1.2	0.21	−0.1	−0.06
Random forest, re-tuned daily	0.1	−0.15	0.2	0.10
LightGBM, re-tuned every 25 days	−0.5	0.12	1.1	0.30
XGBoost, re-tuned every 25 days	0.2	0.19	−0.3	−0.05
Random forest, re-tuned every 25 days	−0.3	0.16	0.2	0.12
Panel B: Full predictor set				
LightGBM, re-tuned daily	0.1	−0.06	0.4	−0.02
XGBoost, re-tuned daily	−1.3	−0.30	1.7	0.07
Random forest, re-tuned daily	0.5	−0.80	0.7	0.02
LightGBM, re-tuned every 25 days	0.9	0.28	3.8	0.70
XGBoost, re-tuned every 25 days	−3.7	−0.25	0.6	0.00
Random forest, re-tuned every 25 days	3.2	−0.56	1.5	0.22

Notes: Columns: the best of 50 trials minus the best of the first 10 or 25 trials; rows: Optuna re-tuned daily or every 25 trading days; every specification is re-estimated daily (Section 5.4). Cells as in Table 20.

Table 25: Optuna against simpler tuning and the closing-interval ridge

Re-tuning		ΔQLIKE		ΔSR, mid-quotes	
frequency (trading days)	Specifications	< 0	> 0	< 0	> 0
Panel A: Optuna minus fixed hyperparameters, both re-estimated daily					
1	6	0 (0)	6 (3)	4 (0)	2 (0)
5	6	0 (0)	6 (3)	4 (0)	2 (0)
25	6	0 (0)	6 (4)	2 (0)	4 (0)
250	6	0 (0)	6 (3)	1 (0)	5 (0)
All	24	0 (0)	24 (13)	11 (0)	13 (0)
Panel B: Optuna minus random search, both re-estimated daily					
1	6	4 (0)	2 (0)	4 (0)	2 (0)
5	6	5 (0)	1 (0)	5 (0)	1 (0)
25	6	4 (0)	2 (1)	3 (0)	3 (0)
250	6	3 (0)	3 (0)	4 (0)	2 (1)
All	24	16 (0)	8 (1)	16 (0)	8 (1)
Panel C: Optuna minus the closing-interval ridge					
1	6	0 (0)	6 (3)	6 (1)	0 (0)
5	6	0 (0)	6 (3)	6 (0)	0 (0)
25	6	0 (0)	6 (3)	5 (0)	1 (0)
250	6	0 (0)	6 (3)	5 (0)	1 (0)
All	24	0 (0)	24 (12)	22 (1)	2 (0)

Notes: Optuna (best of 50 trials, re-tuned every 1, 5, 25 or 250 trading days) minus the comparator; specifications: tree model × predictor set. Each cell counts the specifications whose difference lies below or above zero; in parentheses, the number significant at the 5% level. QLIKE: Diebold–Mariano test with the Harvey–Leybourne–Newbold correction (Newey–West lag $q_{\mathrm{NW}} = 6$); Sharpe ratio at mid-quotes: HAC t-statistic on the daily return differential (Newey–West, $q_{\mathrm{NW}} = 14$). The *All* rows pool the four re-tuning frequencies (Section 5.4).

Table 26: Optuna-selected hyperparameters within the search space

		Lower end (%)			Upper end (%)			Median selection	
Hyperparameter	Search range	HAR	Full	Prior	HAR	Full	Prior	HAR	Full
Panel A: LightGBM									
Maximum number of leaves	[2, 128], log, integer	20.6	5.7	11.1	2.0	3.3	4.8	6	15
Minimum observations per leaf	[1, 256], log, integer	19.1	14.2	18.3	0.7	0.8	4.2	5	12
Share of predictors per tree	[0.05, 1]	1.4	5.0	5.2	7.9	5.3	5.3	0.646	0.553
Share of observations per tree	[0.3, 1]	5.2	4.1	4.7	5.0	4.6	4.9	0.638	0.666
L_2 penalty λ_2^{tree}	$[10^{-3}, 10^4]$, log	5.4	5.6	5.5	0.0	0.0	5.1	0.633	0.637
Learning rate η	[0.005, 0.1], log	3.1	3.7	4.9	6.3	4.7	5.2	0.0256	0.0203
Boosting rounds $B^\dagger$	$B \leq \bar{B}(\eta)$	–	–	–	1.1	0.6	16.9–17.5	272	294
Panel B: XGBoost									
Maximum depth	[1, 8], integer	18.4	6.1	12.2	5.3	10.6	12.5	3	5
Minimum Hessian weight per leaf	[0.1, 256], log	6.3	4.4	5.3	0.7	0.8	4.8	2.24	5.88
Share of observations per tree	[0.15, 1]	7.5	3.6	5.2	3.8	5.1	5.3	0.516	0.598
Share of predictors per tree	[0.05, 1]	0.8	4.3	4.7	8.3	5.6	4.9	0.691	0.569
L_2 penalty λ_2^{tree}	$[10^{-5}, 10^4]$, log	4.6	4.2	5.5	0.0	0.0	5.1	0.122	0.0899
Learning rate η	[0.005, 0.1], log	3.9	5.6	4.9	4.2	3.9	5.2	0.024	0.0189
Boosting rounds $B^\dagger$	$B \leq \bar{B}(\eta)$	–	–	–	1.6	1.1	14.5–15.6	298	316
Panel C: Random forest									
Share of predictors per split	[0.02, 1]	0.1	1.2	5.1	6.4	10.3	4.8	0.612	0.452
Minimum observations per leaf	[1, 256], log, integer	31.7	19.1	17.7	0.0	0.0	4.4	2	4
Maximum depth	{3, 5, 10, 20, 40, none}	4.2	3.9	17.0	19.5	22.6	16.5	20	20

Notes: The share of best-of-50 selections in the outermost twentieth of each axis, at its lower and upper end, by predictor set (HAR: HAR benchmark; Full: full predictor set), alongside the share of the random start-up draws in the same region (prior), and the median selection. Each tree model's axes are its own search space (Section 5.3; Table 5); log: log scale; integer: integer-valued. †Not searched directly: boosting rounds B, capped at $\bar{B}(\eta)$.

Table 27: Development of the closing-interval MinGRU network

	HAR benchmark			Full predictor set		
Specification	QLIKE	Δ (%)	SR	QLIKE	Δ (%)	SR
Panel A: Closing-interval models						
(1) Softplus output, MSE	0.1027	6.2***	1.29	0.1108	15.6***	1.20
(2) + encoder weight decay ν_{enc}	0.1015	5.0**	1.49	0.1078	12.5**	1.47
(3) + linear shortcut, zero-initialized	0.0998	3.2	1.34	0.1193	24.5**	1.30
(4) + shortcut ridge-initialized	0.1009	4.4*	1.47	0.1046	9.2**	0.78
(5) (4), selected on QLIKE	0.0987	2.1	1.68	0.1000	4.3	1.67
(6) (4), trained and selected on QLIKE	0.1006	4.1*	1.04	0.0966	0.8	1.53
Panel B: Pooled intraday models, evaluated at the closing interval						
(7) Weight decay ν, selected per block	0.1164	20.4***	1.04	0.1201	25.3***	1.83
(8) As specification (6)	0.1177	21.7***	0.78	0.1141	19.1***	0.34
Panel C: Closing-interval linear models						
Elastic net (comparator)	0.0967	–	1.37	0.0958	–	1.92
Ridge	0.0967	0.0	1.37	0.0975	1.7	1.42

Notes: Each row is a rolling-window out-of-sample exercise over the 866 trading days of the out-of-sample period. Δ (%): $100\,(\overline{\text{QLIKE}}/\overline{\text{QLIKE}}_{\text{enet}} - 1)$ relative to the closing-interval elastic net on the same predictor set; SR: annualized Sharpe ratio of the sign strategy at mid-quotes. *, ** and *** denote significance of Δ at the 10%, 5% and 1% levels (Diebold–Mariano test, Harvey–Leybourne–Newbold correction, Newey–West lag $q_{\text{NW}} = 6$; not computed for the ridge; –: the comparator). Rows (2)–(4) each add the change in their label to the row above, (5) and (6) modify (4), and (8) is the pooled intraday analogue of (6); new penalties are selected at each of the 5 block starts (Section 5.2). The mean QLIKE over all intraday intervals is 0.2298 (full predictor set) and 0.2358 (HAR benchmark) for (8), against 0.2630 and 0.2480 for (7).

Table 28: All estimated MinGRU specifications

	HAR benchmark			Full predictor set		
Specification	QLIKE	Δ (%)	SR	QLIKE	Δ (%)	SR
Panel A: Output link						
Identity	0.1038	7.3***	0.89	0.1293	35.0*	0.84
Softplus [row (1)]	0.1027	6.2***	1.29	0.1108	15.6***	1.20
Panel B: Weight decay ν on all weights						
$\nu = 0.01$	0.1034	7.0***	1.33	0.1243	29.7**	0.85
$\nu = 0.03$	0.1013	4.8**	1.29	0.1124	17.3***	1.43
$\nu = 0.1$	0.1044	8.0***	1.54	0.1086	13.3***	1.20
$\nu = 0.3$	0.1015	5.0**	1.12	0.1132	18.1***	0.99
$\nu = 1$	0.1005	4.0*	1.48	0.1099	14.7***	1.33
ν selected per block	0.1027	6.2***	1.29	0.1247	30.2**	0.76
Panel C: Encoder weight decay ν_{enc}						
$\nu_{\text{enc}} = 0.03$	–	–	–	0.1139	18.9***	1.47
$\nu_{\text{enc}} = 0.3$	–	–	–	0.1089	13.6***	0.80
$\nu_{\text{enc}} = 1$	–	–	–	0.1121	17.0**	0.90
$\nu_{\text{enc}} = 3$	–	–	–	0.1089	13.7**	1.05
$\nu_{\text{enc}} = 6$	–	–	–	0.1074	12.1***	1.28
$\nu_{\text{enc}} = 15$	–	–	–	0.1037	8.2**	0.60
ν_{enc} selected per block [row (2)]	0.1015	5.0**	1.49	0.1078	12.5**	1.47
Panel D: Linear shortcut, zero-initialized						
Shortcut only ($f^{\text{NN}} \equiv 0$)	–	–	–	0.1058	10.4***	1.53
$\nu_{\text{enc}} = 0$	–	–	–	0.1342	40.1***	0.82
$\nu_{\text{enc}} = 15$	–	–	–	0.1173	22.4**	1.51
Penalty selected per block [row (3)]	0.0998	3.2	1.34	0.1193	24.5**	1.30
Panel E: Linear shortcut, ridge-initialized						
Check: ridge shortcut only, identity output	–	–	–	0.0975	1.7	1.42
Check: ridge shortcut only, softplus output	–	–	–	0.0982	2.5	1.78
$\nu_{\text{enc}} = 15$	–	–	–	0.1097	14.5*	1.45
Penalty selected per block [row (4)]	0.1009	4.4*	1.47	0.1046	9.2**	0.78
Panel F: QLIKE in selection and training						
(1), selected on QLIKE	0.1022	5.7***	1.34	0.1108	15.6***	1.20
(2), selected on QLIKE	0.1017	5.2**	1.27	0.1049	9.5**	1.28
(4), selected on QLIKE [row (5)]	0.0987	2.1	1.68	0.1000	4.3	1.67
(4), trained and selected on QLIKE [row (6)]	0.1006	4.1*	1.04	0.0966	0.8	1.53
Panel G: Pooled intraday models, evaluated at the closing interval						
$\nu = 0$	0.1193	23.4***	0.98	0.1273	32.8***	1.90
$\nu = 0.01$	0.1197	23.8***	1.01	0.1218	27.1**	2.23
$\nu = 0.03$	0.1168	20.8***	0.92	0.1204	25.6**	1.84
$\nu = 0.1$	0.1168	20.9***	1.33	0.1200	25.3**	2.30
$\nu = 0.3$	0.1166	20.6***	0.88	0.1175	22.6***	1.73
$\nu = 1$	0.1177	21.7***	1.15	0.1125	17.4***	1.58
ν selected per block [row (7)]	0.1164	20.4***	1.04	0.1201	25.3***	1.83
Ridge-initialized shortcut, QLIKE [row (8)]	0.1177	21.7***	0.78	0.1141	19.1***	0.34

Notes: 35 variants, grouped by the modification each tests; columns, stars and comparator as in Table 27, whose rows are (1)–(8). ν: uniform weight decay on every weight; ν_{enc}: weight decay on the encoder's input columns only; a fixed value is held in every block, or the value is selected per block; "[row (k)]" marks row (k) of Table 27. Shortcut only ($f^{\text{NN}} \equiv 0$): the recurrent branch is held at zero, so the forecast is the linear shortcut alone; ridge shortcut only, identity / softplus output: checks with the shortcut held at the linear models' ridge estimate. –: estimated on the full predictor set only.

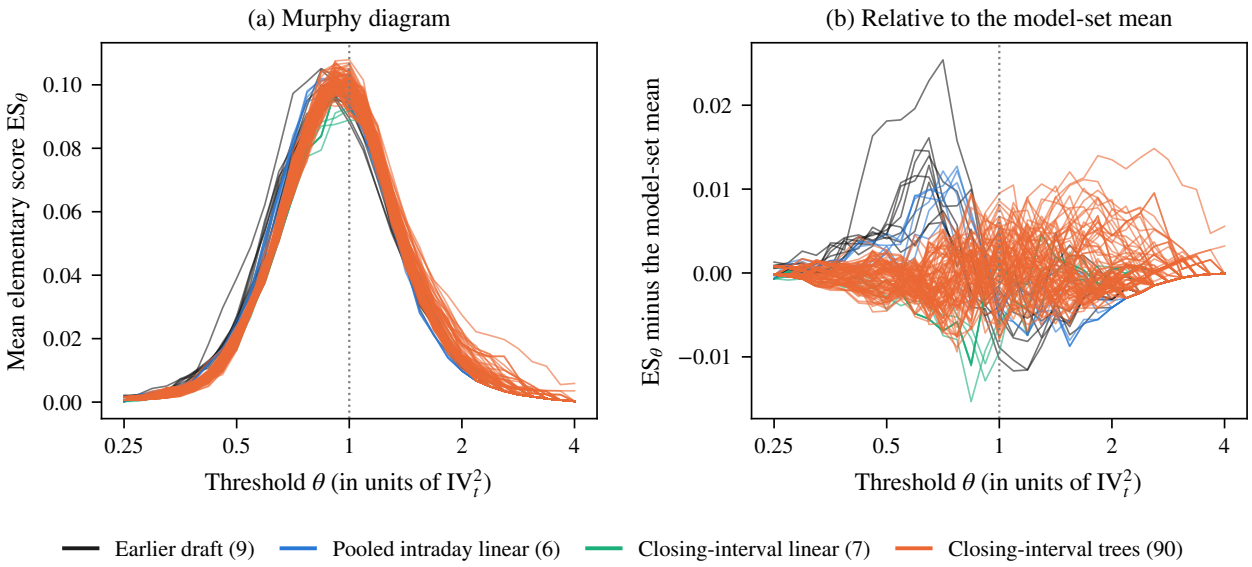


Figure 4: Murphy diagram (elementary scores) for the model set (112 forecasts: earlier draft 9, pooled intraday linear 6, closing-interval linear 7, closing-interval trees 90), over the 866 trading days of the out-of-sample period. Panel (a): mean elementary score $\overline{\mathrm{ES}}_\theta$ (Ehm et al., 2016) at each of 33 thresholds θ, in units of the 15:30 implied variance, log scale (lower is better); $\theta = 1$ is the threshold used by the sign strategy, $q_t = \mathrm{sign}(\hat{\sigma}_t^2 - \mathrm{IV}_t^2)$, and QLIKE equals the integral of the elementary scores weighted by θ^{-2} (Section 7.2). Panel (b): each forecast's mean elementary score minus the model-set mean at the same θ.

Table 29: MinGRU accuracy by type of trading day

Specification	Month-ends (43 days)	Extreme predictors (60 days)	Other days (763 days)	All days (866 days)
Panel A: HAR benchmark				
(1) Softplus output, MSE	0.1672	0.1328	0.0967	0.1027
(2) + encoder weight decay ν_{enc}	0.1722	0.1091	0.0969	0.1015
(3) + linear shortcut, zero-initialized	0.1637	0.0902	0.0969	0.0998
(4) + shortcut ridge-initialized	0.1598	0.1136	0.0966	0.1009
(5) (4), selected on QLIKE	0.1524	0.1062	0.0951	0.0987
(6) (4), trained and selected on QLIKE	0.1411	0.1190	0.0969	0.1006
Elastic net	0.1743	0.1163	0.0907	0.0967
Ridge	0.1743	0.1163	0.0908	0.0967
Panel B: Full predictor set				
(1) Softplus output, MSE	0.2662	0.2165	0.0938	0.1108
(2) + encoder weight decay ν_{enc}	0.2216	0.1735	0.0962	0.1078
(3) + linear shortcut, zero-initialized	0.1626	0.3316	0.1001	0.1193
(4) + shortcut ridge-initialized	0.1856	0.1818	0.0940	0.1046
(5) (4), selected on QLIKE	0.1467	0.1362	0.0945	0.1000
(6) (4), trained and selected on QLIKE	0.1401	0.1527	0.0898	0.0966
Elastic net	0.1557	0.1764	0.0861	0.0958
Ridge	0.1556	0.1852	0.0873	0.0975
Panel C: Full predictor set minus HAR benchmark				
(1) Softplus output, MSE	0.0991	0.0837	−0.0029	0.0082
(2) + encoder weight decay ν_{enc}	0.0494	0.0644	−0.0007	0.0063
(3) + linear shortcut, zero-initialized	−0.0011	0.2414	0.0032	0.0195
(4) + shortcut ridge-initialized	0.0258	0.0682	−0.0026	0.0037
(5) (4), selected on QLIKE	−0.0057	0.0299	−0.0006	0.0013
(6) (4), trained and selected on QLIKE	−0.0011	0.0337	−0.0071	−0.0040
Elastic net	−0.0185	0.0601	−0.0046	−0.0008
Ridge	−0.0187	0.0689	−0.0034	0.0008

Notes: Mean QLIKE of specifications (1)–(6) of Table 27, and of the closing-interval elastic net and ridge, by day type. Month-ends: the last trading day of each calendar month (5 of them also have extreme predictor values); extreme predictor values: at least 10 retained columns above their estimation-window maximum, month-ends excluded. The three day types partition the 866 trading days of the out-of-sample period, so their day-weighted mean equals the overall mean QLIKE. Of the summed daily QLIKE loss differential of specification (1) between the full predictor set and the HAR benchmark, days with extreme predictor values account for 71%, month-ends for 60% and the other days for −31%.

Table 30: Predictor definitions.

Variable	Description	Source	Periodicity adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
Realized moments (7)					
r	Interval log return, $r_{t,i} = \sum_k r_{t,i,k}$	ES futures	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
RAV	Realized absolute variation, $\sum_k \lvert r_{t,i,k}\rvert$	ES futures	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
RSK	Realized skewness, unnormalized, $\sum_k r_{t,i,k}^3$	ES futures	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
RQ	Realized quarticity, unscaled, $\sum_k r_{t,i,k}^4$	ES futures	mean	$\text{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
RS^+	Positive realized semivariance, $\sum_k r_{t,i,k}^2 \mathbf{1}\{r_{t,i,k} > 0\}$	ES futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
BV	Bipower variation, $\text{BV}_{t,i} = \sum_{k\geq 2} \lvert r_{t,i,k}\rvert\,\lvert r_{t,i,k-1}\rvert$	ES futures	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
AC	First-order realized autocovariance, $\text{AC}_{t,i} = \sum_{k\geq 2} r_{t,i,k}\, r_{t,i,k-1}$	ES futures	s.d.	$\text{sign}(v)\,\lvert v\rvert^{1/2}$	$u^{(k)}$, $\bar{a}^{(k)}$
Liquidity (14)					
Volume	ES contract volume	ES futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
N	Number of one-minute returns, $N_{t,i} \leq 30$; mode 30 on 72% of intervals	ES futures	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
TO^{EW}	Equal-weighted turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{EW}}_{\text{buy}}$	Equal-weighted buyer-initiated turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{EW}}_{\text{sell}}$	Equal-weighted seller-initiated turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{EffSpread}^{\text{EW}}$	Equal-weighted effective spread	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{Spread}^{\text{EW}}$	Equal-weighted quoted spread	Constituents	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
TO^{VW}	Value-weighted turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{VW}}_{\text{buy}}$	Value-weighted buyer-initiated turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{TO}^{\text{VW}}_{\text{sell}}$	Value-weighted seller-initiated turnover	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{EffSpread}^{\text{VW}}$	Value-weighted effective spread	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{Spread}^{\text{VW}}$	Value-weighted quoted spread	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
OFI^{EW}	Equal-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
OFI^{VW}	Value-weighted order-flow imbalance, (Buy − Sell)/(Buy + Sell) of turnover	Constituents	s.d.	v	$u^{(k)}$, $\bar{a}^{(k)}$
Equal-weighted market (6)					
RV^{EW}	Equal-weighted realized variance	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
RSK^{EW}	Equal-weighted realized skewness (unnormalized)	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$

continued on the next page

Table 30 (continued)

Variable	Description	Source	Periodicity adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
$\mathrm{RQ}^{\mathrm{EW}}$	Equal-weighted realized quarticity (unscaled)	Constituents	mean	$\mathrm{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RAV}^{\mathrm{EW}}$	Equal-weighted realized absolute variation	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{BV}^{\mathrm{EW}}$	Equal-weighted bipower variation	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RS}^{+,\mathrm{EW}}$	Equal-weighted positive realized semivariance	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Value-weighted market (6)					
$\mathrm{RV}^{\mathrm{VW}}$	Value-weighted realized variance	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RSK}^{\mathrm{VW}}$	Value-weighted realized skewness (unnormalized)	Constituents	s.d.	$v^{1/3}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RQ}^{\mathrm{VW}}$	Value-weighted realized quarticity (unscaled)	Constituents	mean	$\mathrm{sign}(v)\,\lvert v\rvert^{1/4}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RAV}^{\mathrm{VW}}$	Value-weighted realized absolute variation	Constituents	mean	v	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{BV}^{\mathrm{VW}}$	Value-weighted bipower variation	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
$\mathrm{RS}^{+,\mathrm{VW}}$	Value-weighted positive realized semivariance	Constituents	mean	$\sqrt{v}$	$u^{(k)}$, $\bar{a}^{(k)}$
Sentiment (3)					
Attn	Message volume (attention)	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Sent	Net sentiment score	StockTwits	none	v	$u^{(k)}$, $\bar{a}^{(k)}$
NSent	Number of sentiment-tagged messages	StockTwits	mean	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Implied volatility (3)					
VIX	VIX index (30-day S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
VVIX	VVIX index (implied volatility of VIX)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
VIX3M	VIX3M index (3-month S&P 500 implied volatility)	Cboe	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
Volatility demand (4)					
$\mathrm{VD}_{\mathrm{OC}}^{\mathrm{SPX}}$	Net volatility demand, SPX options, opening and closing trades; mode 0 on 59% of intervals	Cboe	none	$\mathrm{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{O}}^{\mathrm{SPX}}$	Net volatility demand, SPX options, opening trades; mode 0 on 60% of intervals	Cboe	none	$\mathrm{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{OC}}^{\mathrm{All}}$	Net volatility demand, all index options, opening and closing trades; mode 0 on 59% of intervals	Cboe	none	$\mathrm{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\mathrm{VD}_{\mathrm{O}}^{\mathrm{All}}$	Net volatility demand, all index options, opening trades; mode 0 on 60% of intervals	Cboe	none	$\mathrm{asinh}(u/\kappa)$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
FOMC (4; closing-interval model only)					
$\mathrm{FOMC}^{\mathrm{rel}}$	FOMC announcement-interval dummy (14:30 ET); mode 0 on 98% of intervals	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$

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Table 30 (continued)

Variable	Description	Source	Periodicity adjustment	Transform $g(v)$	Predictors, $k \in \mathcal{K}$
FOMC^{day}	FOMC announcement-day dummy; mode 0 on 95% of intervals	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$, $\bar{\chi}^{(k)}$
$\text{FOMC}^{\text{next}}$	$1/(1+n^{+}_{t,i})$, $n^{+}_{t,i}$ intervals to the next announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$
$\text{FOMC}^{\text{last}}$	$1/(1+n^{-}_{t,i})$, $n^{-}_{t,i}$ intervals since the last announcement	FOMC calendar	none	$\log v$	$u^{(k)}$, $\bar{a}^{(k)}$

Notes: Each series u enters as $u^{(k)}$, its adjusted, transformed and winsorized value averaged over the previous k intervals, and as $\bar{a}^{(k)}(u)$, the share of those intervals on which it was observed; series with an activity indicator also enter as $\bar{\chi}^{(k)}(u)$, the share off their mode (the mode and its share of intervals are in the description). $\mathcal{K} = \{1, 5, 25, 125, 625, 3125\}$ in the closing-interval model and $\{1, 2, 4, \ldots, 2048\}$ in the pooled intraday model, which omits the FOMC group. Periodicity adjustment: division by the trailing same-interval mean (mean) or standard deviation (s.d.); g: the variance-stabilizing map; the volatility-demand series take $\operatorname{asinh}(u_{t,i}/\kappa_{t,i})$ in place of both (Section 4). A series is signed if it has any negative value: $\text{Spread}^{\text{VW}}$ has a single negative value (2018-11-12 11:00 ET) and so takes $g(v) = v$ and the s.d. adjustment, where $\text{Spread}^{\text{EW}}$ takes $\log v$ and the mean. Unnormalized and unscaled: without the factors $\sqrt{N_{t,i}}/\text{RV}_{t,i}^{3/2}$ (skewness), $N_{t,i}/3$ (quarticity) and $\pi/2$ (bipower variation).

Table 31: Ranking of all forecasts by mean QLIKE

					vs. HAR benchmark	
Rank	Model	Specification	Predictors	QLIKE	Δ (%)	DM
Panel A: Closing-interval models						
1	Lasso	–	Full	0.0931	−3.7	(−1.35)
2	XGBoost	Fixed, daily	Full	0.0957	−1.0	(−0.31)
3	Elastic net	–	Full	0.0958	−0.9	(−0.21)
4	LightGBM	Fixed, daily	Full	0.0961	−0.6	(−0.16)
5	MinGRU‡	–	Full	0.0966	−0.1	(−0.02)
6	Elastic net	–	HAR	0.0967	0.0	(−1.15)
7	Lasso	–	HAR	0.0967	0.0	(−1.38)
8	Ridge	–	HAR	0.0967	0.0	(−0.34)
9	OLS (benchmark)	–	HAR	0.0967	–	–
10	Ridge	–	Full	0.0975	0.8	(0.18)
11	LightGBM	Random search (QLIKE), daily	Full	0.0980	1.4	(0.38)
12	LightGBM	Fixed, every 10 d	Full	0.0983	1.7	(0.47)
13	Random forest	Random search (QLIKE), every 10 d	HAR	0.0989	2.3	(1.11)
14	XGBoost	Fixed, every 10 d	Full	0.0990	2.4	(0.63)
15	LightGBM	Random search (MSE), daily	Full	0.0990	2.4	(0.62)
16	Random forest	Fixed, daily	Full	0.0991	2.5	(0.71)
17	Random forest	Fixed, daily	HAR	0.0992	2.6	(1.23)
18	Random forest	Random search (QLIKE), daily	HAR	0.0995	3.0	(1.39)
19	LightGBM	Random search (QLIKE), every 10 d	Full	0.0998	3.2	(0.78)
20	XGBoost	Fixed, every 10 d	HAR	0.0999	3.3	(1.51)
21	XGBoost	Optuna (MSE), daily, 25 trials	Full	0.1000	3.4	(0.86)
22	XGBoost	Fixed, daily	HAR	0.1001	3.5	(1.53)
23	Random forest	Random search (QLIKE), daily	Full	0.1002	3.7	(1.03)
24	LightGBM	Random search (MSE), every 10 d	Full	0.1005	4.0	(0.90)
25	XGBoost	Optuna (MSE), 5 d, 50 trials	Full	0.1005	4.0	(1.05)
26	Random forest	Optuna (MSE), daily, 25 trials	Full	0.1006	4.0	(1.09)
27	Random forest	Optuna (MSE), 25 d, 10 trials	Full	0.1006	4.0	(1.05)
28	Random forest	Fixed, every 10 d	HAR	0.1006	4.0*	(1.90)
29	Random forest	Fixed, every 10 d	Full	0.1006	4.0	(1.16)
30	MinGRU‡	–	HAR	0.1006	4.1*	(1.72)
31	LightGBM	Optuna (MSE), 250 d, 50 trials	Full	0.1006	4.1	(0.92)
32	Random forest	Optuna (MSE), daily, 10 trials	Full	0.1008	4.2	(1.13)
33	Random forest	Optuna (QLIKE), 25 d, 50 trials	HAR	0.1008	4.2*	(1.90)
34	XGBoost	Random search (MSE), daily	Full	0.1009	4.4	(1.02)
35	XGBoost	Random search (QLIKE), daily	Full	0.1009	4.4	(1.04)
36	LightGBM	Fixed, daily	HAR	0.1010	4.5**	(2.04)
37	Random forest	Optuna (MSE), 5 d, 50 trials	Full	0.1012	4.6	(1.17)
38	Random forest	Optuna (MSE), daily, 50 trials	Full	0.1012	4.7	(1.25)
39	XGBoost	Optuna (MSE), 250 d, 50 trials	Full	0.1013	4.7	(0.98)
40	Random forest	Optuna (MSE), daily, 25 trials	HAR	0.1013	4.8**	(2.16)
41	Random forest	Optuna (MSE), daily, 10 trials	HAR	0.1014	4.9**	(2.18)
42	LightGBM	Optuna (MSE), 5 d, 50 trials	Full	0.1014	4.9	(1.23)
43	Random forest	Optuna (MSE), 5 d, 50 trials	HAR	0.1015	4.9**	(2.15)
44	Random forest	Optuna (MSE), daily, 50 trials	HAR	0.1015	5.0**	(2.07)
45	Random forest	Optuna (MSE), 25 d, 25 trials	HAR	0.1015	5.0**	(2.19)
46	XGBoost	Optuna (MSE), daily, 50 trials	Full	0.1017	5.2	(1.36)

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Table 31, continued

					vs. HAR benchmark	
Rank	Model	Specification	Predictors	QLIKE	Δ (%)	DM
47	Random forest	Optuna (MSE), 25 d, 50 trials	HAR	0.1017	5.2**	(2.34)
48	LightGBM	Fixed, every 10 d	HAR	0.1018	5.3**	(2.27)
49	Random forest	Optuna (MSE), 25 d, 10 trials	HAR	0.1020	5.5**	(2.26)
50	Random forest	Optuna (MSE), 25 d, 25 trials	Full	0.1022	5.8	(1.53)
51	XGBoost	Optuna (MSE), 5 d, 50 trials	HAR	0.1023	5.8**	(2.16)
52	Random forest	Optuna (QLIKE), 25 d, 50 trials	Full	0.1023	5.9	(1.45)
53	XGBoost	Optuna (MSE), daily, 10 trials	HAR	0.1024	5.9**	(2.27)
54	LightGBM	Optuna (MSE), 25 d, 25 trials	Full	0.1026	6.1	(1.35)
55	LightGBM	Random search (QLIKE), daily	HAR	0.1026	6.2***	(2.70)
56	Random forest	Optuna (MSE), 250 d, 50 trials	Full	0.1027	6.3*	(1.70)
57	LightGBM	Random search (QLIKE), every 10 d	HAR	0.1027	6.3***	(2.77)
58	LightGBM	Optuna (MSE), daily, 25 trials	Full	0.1028	6.3	(1.59)
59	Random forest	Random search (QLIKE), every 10 d	Full	0.1029	6.4*	(1.69)
60	XGBoost	Random search (QLIKE), every 10 d	Full	0.1029	6.5	(1.42)
61	XGBoost	Random search (MSE), every 10 d	Full	0.1030	6.5	(1.44)
62	XGBoost	Optuna (MSE), 25 d, 10 trials	HAR	0.1030	6.6**	(2.56)
63	XGBoost	Optuna (MSE), 25 d, 25 trials	Full	0.1031	6.6	(1.46)
64	XGBoost	Optuna (MSE), daily, 10 trials	Full	0.1031	6.6*	(1.68)
65	LightGBM	Optuna (MSE), daily, 10 trials	Full	0.1031	6.6	(1.53)
66	XGBoost	Optuna (MSE), 25 d, 50 trials	HAR	0.1032	6.7**	(2.37)
67	LightGBM	Optuna (MSE), daily, 50 trials	Full	0.1032	6.7*	(1.65)
68	XGBoost	Optuna (QLIKE), 25 d, 50 trials	HAR	0.1034	7.0*	(1.93)
69	XGBoost	Optuna (MSE), 25 d, 25 trials	HAR	0.1035	7.1**	(2.38)
70	LightGBM	Optuna (QLIKE), 25 d, 50 trials	HAR	0.1036	7.1***	(2.79)
71	XGBoost	Optuna (MSE), daily, 50 trials	HAR	0.1036	7.2**	(2.41)
72	XGBoost	Optuna (MSE), 25 d, 50 trials	Full	0.1036	7.2	(1.61)
73	XGBoost	Optuna (MSE), daily, 25 trials	HAR	0.1037	7.2**	(2.29)
74	Random forest	Optuna (MSE), 25 d, 50 trials	Full	0.1038	7.3*	(1.84)
75	XGBoost	Optuna (QLIKE), 25 d, 50 trials	Full	0.1039	7.4	(1.56)
77	LightGBM	Optuna (MSE), 25 d, 25 trials	HAR	0.1040	7.6***	(2.90)
78	LightGBM	Optuna (MSE), 5 d, 50 trials	HAR	0.1044	8.0***	(2.98)
79	LightGBM	Optuna (QLIKE), 25 d, 50 trials	Full	0.1045	8.1	(1.60)
80	Random forest	Random search (MSE), daily	Full	0.1045	8.1*	(1.73)
82	LightGBM	Optuna (MSE), daily, 10 trials	HAR	0.1051	8.7***	(3.51)
83	LightGBM	Optuna (MSE), 25 d, 50 trials	HAR	0.1052	8.8***	(3.02)
84	XGBoost	Random search (QLIKE), daily	HAR	0.1052	8.9**	(1.98)
85	LightGBM	Optuna (MSE), daily, 50 trials	HAR	0.1055	9.1***	(3.04)
86	XGBoost	Random search (QLIKE), every 10 d	HAR	0.1055	9.1**	(2.01)
87	LightGBM	Optuna (MSE), 25 d, 10 trials	Full	0.1055	9.1*	(1.82)
88	LightGBM	Optuna (MSE), 25 d, 10 trials	HAR	0.1058	9.4***	(3.21)
89	LightGBM	Optuna (MSE), daily, 25 trials	HAR	0.1058	9.4***	(3.02)
90	Random forest	Optuna (MSE), 250 d, 50 trials	HAR	0.1060	9.7**	(2.25)
91	LightGBM	Optuna (MSE), 25 d, 50 trials	Full	0.1064	10.1**	(2.04)
93	LightGBM	Random search (MSE), every 10 d	HAR	0.1071	10.8***	(2.68)
94	LightGBM	Random search (MSE), daily	HAR	0.1073	11.0***	(2.74)
95	XGBoost	Optuna (MSE), 25 d, 10 trials	Full	0.1076	11.3**	(2.20)
98	Random forest	Random search (MSE), every 10 d	Full	0.1085	12.2*	(1.93)
99	Random forest	Random search (MSE), every 10 d	HAR	0.1089	12.6**	(2.50)
101	Random forest	Random search (MSE), daily	HAR	0.1090	12.8**	(2.56)

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Table 31, continued

Rank	Model	Specification	Predictors	QLIKE	vs. HAR benchmark Δ (%)	DM
103	XGBoost	Optuna (MSE), 250 d, 50 trials	HAR	0.1092	13.0***	(3.14)
107	XGBoost	Random search (MSE), daily	HAR	0.1098	13.6***	(2.72)
109	XGBoost	Random search (MSE), every 10 d	HAR	0.1100	13.7***	(2.83)
110	LightGBM	Optuna (MSE), 250 d, 50 trials	HAR	0.1100	13.7***	(3.06)
Panel B: Pooled intraday models						
76	Elastic net	–	Full	0.1040	7.5	(1.51)
81	Ridge	–	Full	0.1045	8.1*	(1.69)
92	Lasso	–	Full	0.1069	10.6*	(1.96)
100	LightGBM	Earlier draft, no FOMC	Full	0.1090	12.7**	(2.33)
104	Lasso	–	HAR	0.1092	13.0***	(3.39)
105	XGBoost	Earlier draft, no FOMC	Full	0.1092	13.0**	(2.47)
106	Elastic net	–	HAR	0.1096	13.3***	(3.44)
108	Ridge	–	HAR	0.1098	13.6***	(3.37)
112	Random forest†‡	–	Full	0.1103	14.1**	(2.53)
115	MinGRU‡	–	Full	0.1141	18.0***	(4.53)
116	MinGRU‡	–	HAR	0.1177	21.7***	(4.80)
117	XGBoost‡	–	HAR	0.1251	29.4***	(4.90)
118	Random forest‡	–	HAR	0.1269	31.2***	(4.86)
120	LightGBM‡	–	HAR	0.1276	31.9***	(4.85)
Panel C: Pooled intraday models, earlier draft's specification						
96	Lasso	Fixed penalty, no FOMC	Full	0.1078	11.5**	(2.18)
97	Ridge	Block penalty, no FOMC	Full	0.1081	11.9**	(2.24)
102	Ridge	Block penalty	Full	0.1092	13.0**	(2.41)
111	Lasso	Fixed penalty	Full	0.1101	13.9**	(2.57)
113	Lasso	Re-tuned penalty, no FOMC	Full	0.1115	15.4***	(2.69)
114	Elastic net	Re-tuned penalty, no FOMC	Full	0.1134	17.3***	(3.10)
119	OLS	–	HAR	0.1270	31.4***	(4.50)

Notes: Sample, loss, tests and significance markers as in Table 7; Δ and DM compare each forecast with the HAR benchmark, as in Table 9. Specification of the closing-interval trees (Panel A): fixed hyperparameters, random search or Optuna, selecting on MSE or QLIKE, then the re-estimation frequency or, for Optuna, the re-tuning frequency and the number of trials searched; d: trading days (Section 5.4). Panel C holds the earlier draft's pooled intraday linear models, with a block, fixed or re-tuned penalty; "no FOMC" marks the earlier draft's predictor set, which lacks the FOMC predictors. The pooled intraday trees are re-estimated at every interval and the pooled intraday MinGRU monthly (Section 5.1). ‡Not in the model set counted in the text; †preliminary (see Table 7).

B Notes to the authors (not part of the paper)

- **The current title, abstract, introduction and conclusion describe the old 24-hour panel** and need replacing.
- **The results text anchors on the pooled block ridge;** re-base it on the per-bar OLS. Against that OLS, 7 of 111 forecasts have lower QLIKE, none significantly.
- **Data issue to fix before final numbers:**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on all 122 trade days after November 2023;
  - VIX, volume-demand and FOMC feeds end early and are carried forward without limit into 2024.

- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models.
- **Pending:** the pooled random forest (full table) and the pooled ridge-init network.
- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.